

THE INFLUENCE OF PUBLIC, PRIVATE, AND HYBRID FUNDING ON SUSTAINABILITY STARTUP SUCCESS

Master Thesis



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By

Bendix Delfs

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Abstract

This thesis investigates how public, private, and hybrid funding pathways influence the success of sustainable startups. Given that these ventures face unique challenges, such as long development horizons and market failures, the source of their funding is a critical strategic choice. This study examines success across three dimensions: the ability to secure follow-on funding, the likelihood and speed of a commercial exit, and long-term operational survival.

Drawing on a dataset of 1,638 ventures from Crunchbase, a quantitative analysis was performed using logistic regression and Cox proportional hazards models to test the influence of each funding archetype. The findings reveal that the presence of a market-oriented investor, in either a purely private or a hybrid syndicate, is the strongest predictor of achieving market-based success, including securing larger amounts of follow-on funding and accelerating the path to an exit. Conversely, a reliance on public funding is associated with a longer, non-exit-focused trajectory. A critical distinction emerged within public funding: non-dilutive grants act as a powerful certification of quality for future investors, while dilutive public equity can send a negative signal.

The research concludes that a startup's funding choice is a foundational decision that aligns it with a distinct path toward either rapid commercialization or long-term, policy-supported endurance, offering key strategic insights for entrepreneurs and policymakers.

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Artificial Intelligence

During the writing process, AI tools (Gemini, ChatGPT, Copilot) were employed as a productivity aid for language refinement and technical formatting (code and LaTeX). Responsibility for the final work and any errors rests solely with the author.

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Glossary

AIC Akaike Information Criterion.

CoxPH Cox Proportional Hazards.

CSV Comma-Separated Values.

CVC Corporate Venture Capital.

ESG Environmental, Social and Corporate Governance.

HQ Headquarters.

HR Hazard Ratios.

IPO Initial Public Offering.

LRT Likelihood Ratio Test.

NGO Non-Governmental Organization.

OLS Ordinary Least Squares.

OR Odds Ratios.

PE Private Equity.

R&D Research and Development.

RDT Resource Dependency Theory.

VC Venture Capital.

VIF Variance Inflation Factor.

1 Introduction

The global community faces unprecedented environmental and societal challenges, from anthropogenic climate change and species extinction to resource depletion and social inequality (Calvin et al. 2023). In response, there has been a growing recognition of the urgent need for sustainable development - a concept famously defined by the World Commission on Environment and Development (1987) as "development that meets the needs of the present without compromising the ability of future generations to meet their own needs." At its core, this definition presents a twofold challenge: meeting the essential needs of the world's poor while operating within the limitations that current technology and social organization pose on the environment's capacity to provide for present and future generations.

To address these pressing issues, ambitious global goals have been established, including the Sustainable Development Goals (UN General Assembly 2015) and the Paris Agreement (2016), which set clear timeframes and far-reaching targets for sustainable development. Achieving these goals necessitates a profound shift in both consumer and producer behavior (Thøgersen 2021), a societal pivot that is already leading to a greater demand for environmentally friendly products and services, as well as a stronger dedication to sustainable practices (Gast et al. 2017).

While this momentum towards sustainability creates opportunities for established firms to minimize their environmental impact (Gast et al. 2017), the scale and urgency of today's sustainability challenges call for more than incremental change — they demand radical innovation. Consequently, a new generation of sustainable startups is emerging as pivotal actors, devising inventive solutions to address crucial issues like pollution, the depletion of resources, and the propagation of disease (Tiba et al. 2021). These ventures embody the process of sustainable entrepreneurship, defined by Greco and De Jong (2017, p.14) as "the discovery, creation, and exploitation of entrepreneurial opportunities that contribute to sustainability by generating social and environmental gains for others in society."

These ventures exemplify this entrepreneurial model by tying their commercial success directly to positive results for the environment and humanity, which in turn creates value for many different stakeholders (Nunes et al. 2022). Spanning diverse sectors, they often achieve this by translating scientific research into scalable solutions that challenge existing systems and tackle the most pressing societal and environmental issues of our time (Tiba et al. 2021). This distinct focus on impact, amplified by shifting consumer trends and national commitments to a greener economy (Peng and Pan 2024), has fueled a significant increase in the formation and popularity of sustainable startups in recent years (Karani and Mshenga 2021).

However, their crucial role in the sustainable transition (Horne and Fichter 2022) does not shield them from significant and unique hurdles that distinguish their journey from mainstream innovation. Their technologies are often disruptive, requiring expensive, long-horizon development before they can generate revenue. This creates a challenging financial profile characterized by high capital costs, uncertain revenue streams, significant information gaps between founders and financiers, and shifting policy and regulatory barriers (Mukherjee et al. 2024).

These heightened risks and market failures create a difficult funding environment for conventional private investors, who naturally allocate capital where it can most efficiently maximize financial returns. As a result, private funding has historically been insufficient to meet the financing demands of sustainable innovation (Clark et al. 2018). For instance, a 2014 analysis of climate finance revealed that private investment constituted only a fraction of the total capital deployed, even as forecasts predicted trillions of dollars in future climate investment opportunities. This shortfall created a clear and persistent "funding gap" that threatened to stifle the growth of many promising ventures (Clark et al. 2018).

To address this market failure, government intervention became a necessary response. Public funding programs were created with the express purpose of addressing the private market's limitations, investing in underfunded technologies and creating a fairer environment for innovative startups (Berger, Criscuolo, et al. 2025). Through early-stage investment, these public entities send a powerful signal to the private sector; by reducing significant technical risks and offering a credible "technology certification," they actively encourage subsequent private investment (NREL 2003).

This intervention, however, is not always direct. Governments have developed more nuanced, hybrid strategies with the express purpose of "crowding in" private capital (NREL 2003). The core principle is to use public development finance to guide and attract much larger sums of private investment toward areas of public good, such as sustainable technologies (Cull et al. 2024). This strategy has given rise to hybrid funding models, where public entities and private investors co-fund startups. In these arrangements, the government can de-risk projects for private partners by offering subsidized equity, thereby inducing their participation in ventures with high societal benefits (Bai et al. 2021).

The scale of this public-private entanglement is significant. Recent statistics from the European venture capital market have highlighted that government bodies have become a substantial source of capital for Venture Capitalists, at times representing the single largest source of funds raised (Bai et al. 2021; Bertoni and Tykvová 2015).

This deep entanglement of public and private capital has created a complex funding landscape for sustainable startups, which can now follow distinct public, private, or hybrid pathways, each with its own underlying logic and goals

While the need for governmental intervention is clear, its implementation and ultimate efficiency have not escaped scrutiny, sparking a long-standing academic debate most promi-

nently introduced by Lerner 1999. Critics of these programs raise several valid concerns: despite their potential to build innovation ecosystems, they risk displacing private investment, funding projects with limited economic value, or encouraging inefficient practices within the ventures they support (Berger, Criscuolo, et al. 2025; Schulte and Birkenmeier 2024)

However, while much valuable research has assessed this dynamic, it has largely focused on the performance of singular instruments or specific programs. Studies have examined the role of government in venture capital finance (Brander, Du, et al. 2015), the success of public-private syndication deals (Zhang 2018), and the historical performance of state-owned development banks (Gutierrez et al. 2011). This instrument-by-instrument analysis, while essential, can produce a fragmented view, making it difficult to understand the broader strategic implications of a startup's complete funding pathway.

This thesis, therefore, adopts a more holistic and nuanced perspective. First, instead of evaluating isolated instruments, it compares the influence of broad funding archetypes—private, public, and hybrid—to understand how these entire pathways shape startup trajectories. Second, it moves beyond a single definition of success. Acknowledging that a rapid exit may be a triumph for a private investor but a failure for a public agency seeking long-term impact, this research assesses performance across multiple dimensions, including follow-on funding, exit, and long-term survival.

This integrated approach, which respects the distinct mission of each funding model, leads directly to the central research question guiding this thesis:

How do private, public, and hybrid funding archetypes distinctly influence the success of sustainable startups, when success is measured by the likelihood of securing follow-on funding, achieving an exit, and ensuring long-term survival?

To answer this question, the remainder of this thesis is structured as follows. Chapter 2 builds the theoretical foundation. It applies relevant theories to discuss in depth why funding is so critical for sustainable startups, defines the underlying private, public, and hybrid archetypes based on their motivations and the non-financial support they provide, and derives three testable hypotheses from this framework. Chapter 3 outlines the quantitative methodology, introducing the underlying *Crunchbase* dataset, explaining the extensive measures taken to clean and prepare the data for analysis, and detailing the statistical models used to test each hypothesis. Chapter 4 presents the empirical findings, beginning with a descriptive overview of the sample and then sequentially reporting the model results for each of the three hypotheses on follow-on funding, exit, and long-term survival. Finally, Chapter 5 provides a comprehensive discussion that interprets these findings and links them back to the theoretical framework, and Chapter 6 concludes by summarizing the contributions, implications and limitations of the research.

2 The Logics of Funding: A Theoretical Framework for Sustainable Startups

To understand why different funding models might lead to different outcomes, it is essential to first build a robust theoretical framework. This chapter establishes that foundation. It begins by exploring the unique operational and financial challenges inherent to sustainable ventures, using Resource Dependence Theory to explain their profound reliance on external funding. Subsequently, it introduces Institutional Logics to deconstruct the funding landscape into three distinct archetypes — private, public, and hybrid — each governed by different motivations. By analyzing the interplay between startup needs and investor logics, this chapter culminates in a conceptual framework and a set of testable hypotheses that guide the empirical analysis to follow.

2.1 The Unique Challenges of Sustainable Startups

The pursuit of widespread societal benefits exposes sustainable startups to a unique set of hurdles, which can be broadly understood as either external market barriers or internal operational constraints. At the external level, these ventures must often contend with significant market inefficiencies. Many sustainable innovations present a systemic challenge to market economies because they exhibit the characteristics of public goods. Their benefits, which often manifest as improvements to the collective welfare, are non-excludable and non-depletable, placing them in direct contrast with traditional private goods. This is especially true for innovations concerning the environmental commons, where property rights are ill-defined. As a result, market forces alone cannot efficiently allocate resources for their development, leading to their systemic under-provision and a failure to capture their full societal value (Cumming, Henriques, et al. 2016).

Compounding the challenge of market inefficiencies, a limited public understanding of the broader benefits of sustainable products and services can further devalue sustainable business practices. Consumers often struggle to fully appreciate the environmental and societal gains of sustainable offerings compared to traditional alternatives. This necessitates that sustainable businesses frequently invest in educating potential customers about the long-term benefits of their innovations (Gast et al. 2017).

This situation gives rise to a policy paradox for sustainable startups. On one hand, they require an enabling policy environment (streamlined permitting processes or stable carbon pricing), created through targeted interventions, to overcome market inefficiencies and translate their societal benefits into market viability (Temmes et al. 2021). On the other hand, instead of this needed stability, many startups contend with unstable regu-

latory frameworks and frequent policy shifts, which significantly impact financial viability and deter long-term investment (Mukherjee et al. 2024).

Shifting from the external market barriers, the second category of hurdles is rooted in the internal operational and financial profile of sustainable startups themselves. Their ambitious pursuit of disruptive solutions often relies on emerging technologies demanding extensive, long-term investments (Mukherjee et al. 2024). This creates a challenging financial position defined by lengthy development horizons, high upfront capital requirements, and uncertain revenue streams (Kennedy et al. 2024).

These combined factors place startups in the precarious “valley of death” — a critical phase between inception and early commercialization where risk is exceptionally high (Kennedy et al. 2024). In this phase, a knowledge gap between founders and investors creates major hurdles for securing funding (Mukherjee et al. 2024). As a consequence, capital is often allocated to more established, lower-risk infrastructure projects, while transformative green technologies from small and medium-sized businesses remain underfunded (Mukherjee et al. 2024).

Taken together, the external market barriers and internal operational constraints discussed above underscore a profound reliance on outside support for sustainable startups. To navigate this complex operating environment and achieve their dual objectives of commercial viability and societal impact, these ventures depend on a variety of external actors and resources, with funding being particularly critical. This heightened dependency sets the stage for a deeper examination of their strategic interactions with resource providers.

2.2 Theoretical Lens 1: Resource Dependence and the Criticality of Funding

The pronounced challenges faced by sustainable startups highlight their critical need for external resources. This section introduces Resource Dependence Theory (RDT) as a lens to understand these dependencies, with a particular focus on funding as a pivotal resource and investors as key suppliers of this resource.

2.2.1 Strategic Dependencies and the Role of Funding

Resource Dependence Theory, introduced by Pfeffer and Salancik (1978), provides the theoretical lens to understand this dynamic. The theory’s core premise is that organizations are dependent on their environment for critical resources, which influences their behavior and forces them to strategically manage these dependencies to ensure survival and success (Nienhüser 2008; Hillman et al. 2009). This dependence is particularly pertinent for nascent organizations like startups, which must actively acquire resources from their environment to operate and survive (Roundy and Bayer 2019). Their success hinges on effectively managing these external relationships to secure support — which encompasses not only capital but also the specialized knowledge, mentorship, and partnerships

that funding enables (Arrigo et al. 2024) — while failures often stem from an inability to do so (Gautam and Wattanapongsakorn 2024).

Viewing sustainable startups through the lens of RDT reveals a network of critical external dependencies that extend far beyond simple financing needs. Their dual mission—balancing commercial viability with socio-environmental objectives—intensifies this reliance on their external environment. From this RDT perspective, the success of these ventures becomes contingent on their ability to manage a specific set of strategic dependencies:

- *Dependency on the Policy Environment.* The public-good nature of many sustainable innovations creates market failures where their full societal value is not captured by private actors (Cumming, Henriques, et al. 2016). Observed through the lens of RDT, this inherent market limitation, often compounded by unstable regulatory frameworks, translates into a direct dependency on policy actors (Temmes et al. 2021). Navigating this fluctuating policy landscape thus becomes a critical capability for survival and growth (Mukherjee et al. 2024).
- *Dependency on Patient Capital.* The financial profile of sustainable startups, typically defined by long development horizons and high upfront costs, places them in the precarious “valley of death” (Kennedy et al. 2024; Mukherjee et al. 2024). In RDT terms, this is not just a generic need for capital, but a critical dependency on a specific and scarce resource: “patient” investors who are willing to supply financial support under conditions of high risk and delayed returns.
- *Dependency on Legitimacy and Validation.* The significant information asymmetry between founders and financiers presents a major hurdle to securing investment (Karsai 2018). RDT provides a framework for understanding this challenge not merely as a communication problem, but as a critical dependency on acquiring intangible resources. From this viewpoint, gaining legitimacy and third-party validation becomes a strategic imperative to signal the venture’s quality and potential to a skeptical market (Islam et al. 2018; Karsai 2018).

RDT thus frames the journey of a sustainable startup as an exercise in strategically managing the heightened dependencies—on policy, patient capital, and legitimacy—that arise from its dual mission. While startups must secure various resources to navigate this environment, funding stands out as the most pivotal, offering the potential to address these specific dependencies. Therefore, to understand how sustainable ventures can satisfy their complex needs, we must first unpack the nature of funding itself. The following section will do just that, defining funding not merely as capital, but as a critical, multi-dimensional resource.

2.2.2 The Role of Funding as a Critical, Multi-Dimensional Resource

To understand how funding addresses the complex dependencies of sustainable startups, it must be viewed as a multi-dimensional resource package. This package encompasses two primary dimensions. The most evident remains the tangible component: capital. The amount of financial capital a startup secures directly dictates its operational runway and ability to pursue growth, making it a significant predictor of survival for new firms (Coleman et al. 2013). However, this tangible capital is almost always accompanied by a second, intangible dimension comprising non-monetary support mechanisms.

The nature of this intangible support is intrinsically linked to an investor's underlying motivations — be it purely financial gain, a specific policy objective, or a blend of both. These resources are not passive benefits; they are strategic tools actively deployed by investors to steer a venture toward their desired outcomes. This active involvement can be understood through two key facets identified by Hellmann and Puri (2002): first, support, where investors take actions that benefit the company (e.g., aiding in recruitment or strategy), and second, control, where they can influence decisions to maximize the firm's value, even if it conflicts with a founder's preferences.

Viewing this through the lens of RDT highlights why this distinction is so important. Given their heightened dependencies on specific external factors — such as policy stability, patient capital, and market legitimacy — the intangible support a sustainable startup receives is arguably as crucial as the capital itself. This suggests that a venture's success may hinge on its ability to align with funding providers whose strategic support mechanisms are equipped to address these unique challenges. It is therefore essential to examine how different funding institutions vary in the tangible and intangible resources they offer, and consequently, how they are suited to foster the growth of sustainable ventures.

2.3 Theoretical Lens 2: Institutional Logics and Funding Archetypes

The previous section established why sustainable startups have distinct needs and framed funding as the critical, multi-dimensional resource for addressing them. However, the world of funding is not uniform as it encompasses a broad scope of providers with different backgrounds, goals, and investing mechanisms. To understand how these distinct modalities influence startup performance across varied notions of 'success', it is first crucial to map the spectrum of available funding types and the institutional logics that govern them.

To understand why different types of investors operate with such divergent goals, this thesis draws on the concept of institutional logics, which was first introduced by Alford and Friedland in 1985. In this context, an institutional logic can be thought of as the "unwritten rulebook" — the set of socially constructed practices, assumptions, and beliefs—that guides the behavior of actors in a field (Friedland and Alford 1991). In essence, these

frameworks shape what is considered rational and legitimate action for a group (Thornton and Ocasio 2008), whether that group is a government agency or a venture capital fund. While its origins lie in describing broad societal contradictions, institutional logics are now frequently used in funding and investment research to explain the distinct behaviors of market actors (e.g. Pahnke et al. 2015; Xiao and Au 2020). Consequently, the theory provides a fitting framework for analyzing the fundamental differences among the funding archetypes in this study.

Applying the theory of institutional logics to the investment landscape reveals two dominant frameworks that govern the funders central to this thesis: a market logic and a state logic.

The **market logic**, which primarily guides private investors, is rooted in competition and financial return. Under this framework, relationships are shaped by the competitive pursuit of the most promising ventures, and power is determined by market dynamics and corporate control. Legitimacy is conferred almost exclusively through profitability or a clear trajectory towards a lucrative exit; in short, success is defined by maximizing financial value (Thornton and Ocasio 2008).

In contrast, the **state logic** primarily guides public entities and is oriented towards collective benefit and policy achievement. Under this democratic philosophy, the primary concern is contributing to the common good, often by correcting market failures or advancing specific policy goals. Legitimacy is therefore conferred not by profit, but by adherence to public rules and a venture's contribution to broader societal objectives, such as regional development or technological advancement (Thornton, Ocasio, and Lounsbury 2012).

These distinct market and state logics give rise to the three funding archetypes central to this thesis: private, public, and hybrid funding. Each archetype operates with different motivations and, as a result, provides a unique combination of financial capital and the critical non-monetary support mechanisms discussed previously. They can be defined as follows:

- **Private funding.** Guided primarily by a market logic, this category comprises actors whose main motivation is financial return, although an expanding segment pursues explicit impact objectives. It includes venture capital and private equity funds, angel investors, corporations, and dedicated impact-investment vehicles.
- **Public funding.** Guided by a state logic, these investors are typically mission-oriented and deploy resources in line with governmental policy goals. Exemplars include government agencies, national or multilateral development banks, government-owned venture-capital entities, and sovereign-wealth funds with developmental mandates.
- **Hybrid funding.** This archetype represents a formal blend of the two logics, combining public and private resources—through co-investment syndicates or other blended-finance structures—to mobilize private capital toward projects that advance

the public interest or correct market failures.

Having established this framework, the following sections will analyze each archetype in detail. The analysis will focus on their specific motivations, the investment instruments they employ, and the distinct support mechanisms they offer, all viewed through the prism of their governing institutional logics.

2.3.1 Private Funding: Profit-Driven Investment Dynamics

Operating according to the principles of a market logic, private funding represents one of the most significant and diverse sources of external financing available to startups. It refers to investments and support sourced from private-sector entities, including individual investors, private companies, institutional investors, and investment firms (Cumming and Hammer 2025).

Consistent with this logic's focus on competition and profit, investment decisions within private funding are typically mainly driven by profit-oriented objectives, risk-return profiles, and potential strategic alignment. Startups, being inherently uncertain and risky ventures, rely heavily on these external funds to get off the ground. To reward the risk investors, undertake in financing such high-risk projects, they expect a return that appropriately compensates for the uncertainties (Andrieu 2013).

While the primary motivation is financial return, an expanding segment of private capital specifically targets sustainable startups. These so-called impact investors actively seek ventures that can deliver measurable environmental and social gains alongside financial ones. This reflects a more complex motivation where the traditional market logic is blended with principles of social or environmental value creation, aligning more closely with the dual objectives of sustainable entrepreneurs (Trelstad 2016). This shift is not solely fueled by the aim "to do good", but also based on the belief that sustainability can be a vital strategic benefit and can contribute to maximizing the total value of the investments (Bugg-Levine and Emerson 2011).

Investment Instruments of Private Funding

An investment instrument is the specific legal and financial vehicle through which the tangible component of funding — the capital — is provided to a startup. These instruments fundamentally define the startup's obligations and the investor's relationship to the venture. The primary distinction is between equity-based (dilutive) financing, where capital is exchanged for ownership, and debt-based (non-dilutive) financing, where capital is loaned and must be repaid. While early-stage ventures can also rely on less institutionalized sources like personal funds, angel investors, or crowdfunding platforms, this analysis focuses on the more formalized instruments typically employed by institutional investors.

Equity Financing

Equity financing is the most dominant instrument for high-growth startups and the purest

expression of the market logic. It involves exchanging capital for an ownership share (Nazara et al. 2023), a process that directly aligns the interests of investors and founders toward a common goal: increasing the company's value for a lucrative exit event such as an acquisition or initial public offering (IPO) (Metrick and Yasuda 2021). The primary institutional actors employing this model are:

- **Venture Capital (VC) Firms:** Formal investment institutions that manage pooled capital to invest in a portfolio of high-growth startups, acquiring a significant equity stake with the objective of generating high-multiple financial returns for their fund (Metrick and Yasuda 2021).
- **Corporate Venture Capital (CVC):** A specialized arm of a large corporation that makes equity investments to generate financial returns while also gaining strategic advantages for the parent company, such as insight into new technologies or potential acquisitions (Dushnitsky and Lenox 2006).
- **Private Equity (PE) Firms:** Investment firms that typically acquire stakes in more mature companies but may provide "growth equity" to later-stage, proven startups to finance expansion (Gilligan 2021).

Debt Financing

As a form of non-dilutive funding, debt financing involves a loan that must be repaid with interest, meaning the startup does not give up ownership (Nazara et al. 2023). While still driven by a market logic of seeking returns, this instrument is more common for companies at later stages with predictable cash flows, as it is less flexible and exposes the investor to less upside than equity (Oranburg 2020). Unlike equity investors, traditional debt providers like banks do not typically share in the venture's growth.

An increasingly prominent instrument that bridges the gap between debt and equity is convertible debt. This starts as a non-dilutive loan but is designed to convert into equity at a later funding stage, providing flexibility for both startups and investors in valuing an early-stage company (Chaplinsky and Becker 2020).

Finally, it is important to note that an investor's motivation is distinct from their choice of instrument. For instance, impact investments, which specifically target ventures with measurable social and environmental goals, can be deployed through any of these vehicles — equity, debt, or composite finance — with varying expectations for financial returns (Ormiston et al. 2015).

Support Mechanisms of Private Funding

As established through the lens of Resource Dependence Theory, the non-monetary support an investor provides as part of their funding can be as critical to a startup's success as the capital itself. This section examines the primary support mechanisms offered by private funders, viewing them not as incidental benefits, but as the strategic tools through which the market logic is enacted.

For traditional investors, these mechanisms are aimed squarely at de-risking a venture to maximize financial return. For the growing segment of impact investors, however, this support is dual-focused, designed to scale both the company's financial value and its stated sustainability mission. Because this active involvement is most intensive in equity financing, the following discussion adopts a holistic view of the patterns most common to that model. In this context, the role of active investors like venture capitalists extends far beyond that of traditional financial intermediaries, as they contribute significantly to the professionalization and strategic direction of the companies they finance (Hellmann and Puri 2002).

Strategy & Knowledge

To satisfy the market logic's demand for a high-value exit, private investors provide crucial strategic guidance aimed at building a scalable business model. This support often comes in the form of valuable industry knowledge and hands-on advice on how to formulate and execute business strategies, ensuring the startup can effectively compete in the market (Hallen et al. 2014). This guidance is a key tool for de-risking the investment and shaping the startup's path towards a future acquisition or IPO.

Operational Support

Simultaneously, private investors frequently assume a hands-on role in the operations of as startup, contributing to critical management decisions. Represented as members of the board of directors or assisting with recruiting and compensating key personnel (Sahlman 1990), private investors can actively shape the ventures makeup. Moreover, they might help stage funding rounds or structure key business transactions (e.g., mergers and acquisitions), in order to support the startup to scale effectively and influence its potential for a successful exit (Alperovych et al. 2015).

Network and Credibility

The involvement of a well-established private investor brings instant credibility to a startup. The reputation and experience of a private sponsor serve to certify the quality and potential of the venture (Alperovych et al. 2015). This provides what the market logic recognizes as a legitimate 'stamp of approval,' which is vital for attracting future funding rounds and forming strategic partnerships, as it signals to the market that the company has passed rigorous scrutiny (Arrigo et al. 2024).

Furthermore, private investors capitalize extensively on their networks to empower the growth of the ventures they fund. They provide invaluable access to an extensive web of contacts that can open doors to new customers, key partners, and a broader pool of capital from the financial community. Often, this network access directly paves the way for a desired exit event by connecting the startup with potential acquirers (Sahlman 1990).

Sustainability-Specific Guidance

For sustainability-focused investors, conventional support tools are often repurposed to

address a dual objective. Although the methods may appear similar to traditional investing, their application is tailored. For instance, "active ownership" is a common practice, but in this setting, it is specifically leveraged to guide a company's ESG performance (Guterman 2020). This change in focus can also lead to a core adjustment in the investment philosophy itself, as some impact investors deliberately accept different risk-return calculations—such as taking on greater financial risk or lower returns—to achieve a targeted, positive impact (Trelstad 2016).

On a practical level, this specialized support includes several key activities that are unique to the sustainability domain. Investors provide crucial guidance on Impact Measurement and Management, helping the startup define, track, and report on its environmental and social KPIs. They also help with navigating specialized regulations, providing expertise on environmental laws or industry-specific certifications that are vital for market legitimacy (Agrawal and Hockerts 2019; Nachyła and Justo 2024). Finally, they provide access to impact-focused networks, connecting the startup to a specific community of foundations, NGOs, and partners within the sustainable ecosystem (Nachyła and Justo 2024).

This specialized guidance highlights a more complex motivation. The traditional market logic is not abandoned but rather blended with principles of social and environmental value creation, based on the belief that long-term sustainability can itself be a vital strategic benefit that contributes to maximizing the total value of an investment

2.3.2 Public Funding: Policy-driven Investment and Social Objectives

Alongside the successful private funding industry, public funding avenues have emerged as an essential source for early-stage startups in recent years. Guided by a state logic, public funding refers to capital provided by government entities to support entrepreneurial ventures, particularly those that may not attract sufficient private investment due to higher risks or slower financial returns (Schulte and Birkenmeier 2024). The primary goal of this logic is not to capture immediate financial gains, but to achieve broader policy objectives and generate significant positive externalities—societal spillover benefits like cleaner energy or improved public health (Bai et al. 2021).

This focus on societal benefits makes public funding particularly critical for sustainable startups. As established in Section 2.1, the unique challenges these ventures face - such as long investment horizons and high technical risk - often deter private capital (Cumming and Johan 2008; Berger and Hottenrott 2021). These obstacles can create a significant financing gap where private investment alone is insufficient to meet demand. Public funding therefore steps in as a corrective measure, providing the crucial bridge financing needed to help startups achieve initial viability and attract subsequent private investment (NREL 2003). By filling this gap, governments can stimulate innovation, foster regional development, and advance strategic policy goals like decarbonization (Cumming, Knill, et al. 2025; Pierrakis and Saridakis 2019), ultimately facilitating the transition of these startups into scalable businesses that drive long-term economic and technological advancement

(Islam et al. 2018).

Investment Instruments of Public Funding

As with private funding, the investment instruments used by public funders govern the tangible transfer of capital. However, the nature of these instruments is fundamentally different, reflecting the distinct goals of the state logic. In direct alignment with this logic's focus on fostering long-term innovation over seeking immediate financial returns, public funding places a notable emphasis on non-dilutive mechanisms. By providing capital without requiring founders to give up equity, this approach reduces the financial burden on early-stage startups, allowing them to focus on high-risk, long-term R&D. The most common forms are outlined below.

Grants

Grants are arguably the most common form of public support. As a non-dilutive instrument, this funding allows startups to develop and implement their ideas without the pressure of repayment or equity exchange, thus reducing their financial burden (Fini et al. 2023). They are crucial for technology startups facing prohibitively high costs for R&D and prototyping (Okunnuga et al. 2024), allowing them to focus on innovation and long-term vision without worrying about cash flow in the critical early stages (Mueller 2023). This support is substantiated by empirical evidence showing that public grants increase R&D activity and lead to higher innovation output. Importantly, these funds do not "crowd out" private R&D investment but rather complement it, enhancing a firm's overall capacity to conduct research and bring new technologies to market (Errico et al. 2024).

These grants are typically administered by specialized government funding agencies. Operating at federal, state, or local levels, these agencies are an essential pillar for policymakers in almost every developed country, though they vary in size and structure (Ross 2000; Smits and Denis 2014). Their core function is to execute policy by funding research and innovation, and they have increasingly oriented their support towards strategic goals such as advancing sustainable development (Mobjörk and Linnér 2006) and supporting the direct implementation of science into practice (Smits and Denis 2014).

Loans, Subsidies and Tax Credits

These are other forms of primarily non-dilutive support designed to ease the financial burden on startups in capital-intensive or high-risk industries (Mueller 2023) and are often tied to specific goals, such as the aforementioned boost of sustainable technologies or advancement of research and development. Subsidized loans, often offered at below-market interest rates, provide access to capital at a lower cost, while tax credits can significantly reduce R&D expenses (Cull et al. 2024).

Central to the deployment of these instruments is the development bank, a government-backed institution with a mandate to advance economic growth. They often fulfill this mission by providing credit in situations local capital markets prove inadequate. Beyond

delivering the subsidized loans mentioned earlier, these banks often enhance their financial offerings with a wide array of capacity-building and advisory programs, aimed at fostering the development of innovative enterprises that are not adequately served by the private financial system (de Luna-Martinez and Vicente 2012).

Equity Financing

In recent years, some public funders have begun to adopt instruments that mirror those of the private sector, most notably through equity financing. However, even when taking an ownership stake—a hallmark of the market logic—these public entities continue to operate under the principles of the state logic.

Most commonly referred to as Government-backed venture capital, this has become an established way for public capital to be invested in startups. While governments take an ownership stake in the company through these investments, they are not governed by the same pressures as private VCs: Unlike conventional venture capital, they are less constrained by aggressive profit targets, often operate without a specific timeline for divestment, and enjoy freedom from the need to generate rapid returns (Alperovych et al. 2015). In line with the state logic's general goals, these funds therefore favor high-growth startups in strategic sectors like clean energy, healthcare, or technology (Berger, Dechezleprêtre, et al. 2024).

Support Mechanisms of Public Funding

Similar to the discussion of private funding, it is important to acknowledge that the intangible resources associated with public funding vary by instrument. However, to provide a clear, high-level contrast, this section discusses these support mechanisms in a holistic manner.

These mechanisms are best understood as a direct enactment of the state logic. Their primary purpose is often to address the market failures and information asymmetries that can deter private investment. To achieve this, government-backed funding offers more than just capital; it provides startups with valuable credibility and validation, specialized network access and mentorship, all of which are essential for their long-term development and success.

Credibility and Validation

Public funding is uniquely positioned to combat the information asymmetry that deters private investors. Because public funders are guided by policy objectives rather than profit motives, their vetting process is often perceived as impartial and rigorous. Consequently, when a startup secures public capital, particularly from a reputable government program, it receives a powerful signal of credibility. This “stamp of approval” validates the venture's quality and potential for the entire market, including partners and customers (Okunnuga et al. 2024).

The impact of this validation is twofold, providing crucial strategic and operational advan-

tages. Strategically, empirical studies show it often serves as a steppingstone to attract subsequent private venture capital (Islam et al. 2018). Operationally, the enhanced legitimacy confers vital stability and "breathing room," which allows the startup to weather early-stage uncertainty, insulate itself from market pressures, and concentrate on its long-term mission and foundational R&D (Mukherjee et al. 2024).

Network Access and Mentorship

Reflecting the state logic's goal of embedding startups into strategic policy areas, public funding is often accompanied by access to specialized networks and mentorship designed to foster long-term stability.

Instead of focusing solely on commercial connections, these programs often facilitate access to government agencies, policy experts, and industry leaders aligned with public objectives. This support is crucial for navigating complex regulatory landscapes and building the deep institutional relationships necessary for a venture's endurance. Similarly, mentorship initiatives frequently pair founders with experienced professionals who provide strategic counsel (Okunnuga et al. 2024; Mueller 2023). This guidance typically focuses on sustained technological advancement and building a resilient business model, rather than on engineering a rapid exit.

Public Procurement

Another powerful form of non-monetary support is public procurement, where the government acts as both a foundational customer and a resource enabler for a startup. By becoming a crucial first client for new technologies, the government provides a stable and predictable revenue stream, reducing market entry risk and counteracting the market failures associated with public-good innovations (Uyarra et al. 2014). This predictable income provides the financial stability needed for a startup to endure early-stage uncertainty. Simultaneously, by providing access to large-scale public contracts and pooled infrastructure, the government can significantly lower a startup's development costs, further enhancing its operational resilience and capacity for holistic growth (Temmes et al. 2021).

2.3.3 Hybrid Funding: Synthesizing Goals and Mitigating Constraints

Theoretically, hybrid funding represents a formal arena where the previously discussed market and state logics are intentionally blended. It is a structured collaboration between public and private entities that seeks to create a powerful synergy by combining the "best of both worlds": the market logic's discipline and profit-driven efficiency with the state logic's patient, mission-oriented, and de-risking capacity. The rationale for its emergence is best understood by examining the inherent constraints each funding type presents when operating in isolation.

Strict adherence to a market logic provides crucial expertise but can place startups on a 'tight leash' through its intense focus on short-term milestones and financial returns

(Gompers and Lerner 2001). The use of staged capital infusions, while protecting the investor, creates a dependency that can limit a founder's autonomy and strategic flexibility (Sahlman 1990). Furthermore, a startup's reliance on an investor's knowledge and network can lead to conflicts over control or vision (Hallen et al. 2014).

On the other hand, a pure state logic carries its own challenges. Wholly government-backed programs are often hampered by bureaucratic inefficiency, political pressures, and administrative red tape that can significantly delay and complicate the allocation of capital (Brander, Du, et al. 2015; Karsai 2018). Moreover, the public fund managers dispensing this capital are often civil servants whose compensation is not tied to a venture's success, which can lead to a lack of performance-based incentives and urgency compared to their private-sector counterparts (Leleux and Surlemont 2003). These challenges are well-documented in academic literature, which has often questioned the value of state ownership and cited the historically poor performance of some government-led financial institutions (Gutierrez et al. 2011; Alperovych et al. 2015).

By combining these two models, hybrid funding aims to mitigate these respective weaknesses. The public partner's stable, long-term capital can loosen the "tight leash" of purely private financing, giving ventures more breathing room. In turn, the private partner's profit motive and market expertise inject a level of commercial discipline and efficiency that can be lacking in purely public initiatives, addressing the criticism that governments sometimes fail in the task of allocating capital effectively (Alperovych et al. 2015).

This synergistic structure is particularly well-suited for sustainable startups, as their own dual mission—achieving both commercial viability and positive societal impact—is perfectly mirrored by the blended logic of the hybrid model. This synergy is further reinforced by findings that actors in sustainable innovation are themselves increasingly moving toward such hybrid logics (Siefkes et al. 2024). In this ideal arrangement, the private partner validates the business case while the public partner validates the venture's environmental or social mission.

Investment Models in Hybrid Funding

The blending of the market and state logics is realized through several distinct hybrid models. These structures are often more complex than the discrete investment instruments previously discussed, ranging from direct co-investment in a single startup to the institutional design of entire funds and the development of collaborative ecosystems. While diverse in form, these models all share a common strategic goal: using public resources to de-risk opportunities and "crowd in" private-sector capital and expertise

Direct Co-Investment and Syndication

The most direct form of hybrid funding is the co-investment syndicate, where public and private investors invest in the same funding round of a startup. This model is common in both syndicated loans (Fotak and Lee 2020) and venture capital (Bai et al. 2021). A prominent example, particularly in the VC space, is the matching fund model. In this

structure, a public entity, such as a government-backed fund, commits to matching the capital a startup raises from private investors, thereby sharing the risks and due diligence of the investment (Kim et al. 2015).

Hybrid Investment Vehicles

A more structural approach involves creating hybrid investment vehicles, where public and private capital are blended at the fund level. The most prominent example is government-supported venture capital, where a government acts as a Limited Partner in a privately managed VC fund. This model is distinct from fully state-run or government-owned venture capital and is designed to attract further private investment by reducing the fundraising risk for private fund managers (Brander, Du, et al. 2015). The key principle of this hybrid approach remains that while the capital may be partially public - fulfilling the government's aim to complement voids in the private market and foster economic development (Sohn et al. 2012) - the fund's investment decisions and management are left to private professionals, thereby leveraging their market expertise and incentive structures.

Innovation Ecosystems and Networks

Finally, hybrid funding can be fostered at an infrastructural level through collaborative innovation ecosystems. These are often described by the "Triple Helix" model, which emphasizes the interwoven relationships between Academia (universities as sources of research and talent), Industry (private firms and investors as engines of commercialization), and Government (as a provider of funding, stable policy, and regulatory guidance) (Etzkowitz and Leydesdorff 2000). University-affiliated accelerators, public-private research parks, and industry-specific innovation hubs are concrete examples of these ecosystems (Etzkowitz 2008). Within these networks, public agencies, corporations, VCs, and angel investors collaborate to sponsor, mentor, and fund new ventures, creating a fertile ground where syndications and partnerships emerge organically.

Support Mechanisms of Hybrid Funding

The support mechanisms offered by hybrid funding are a direct result of its blended institutional logic, which combines the market-driven discipline of private actors with the policy-oriented resources of public entities. Rather than creating entirely distinct forms of support, hybrid funding's strength lies in the synergy this creates. These synergistic mechanisms can be broadly understood through two complementary dimensions: the structural and financial synergy of the investment itself, and the operational and knowledge synergy that follows

Structural Synergy

The structural synergy of hybrid funding stems from the powerful signal created when public and private entities co-invest. This arrangement creates a form of amplified validation, as the startup has simultaneously earned the confidence of both market- and policy-oriented actors. By passing the due diligence of both a private investor seeking

commercial viability and a public body vetting for technical merit or policy alignment, the startup is significantly de-risked.

This de-risking occurs through tangible structural mechanisms. Public capital, for instance, can be deployed as subsidized equity or through loan guarantees to lower the financial risk for private co-investors (Bai et al. 2021). Critically, the involvement of a private partner simultaneously introduces a strong, market-based incentive structure, ensuring a focus on commercial milestones alongside public policy goals (Brander, Du, et al. 2015). This combination of public de-risking and private incentive alignment results in a venture with a uniquely credible and robust investment profile.

Operational Synergy

Beyond the structural benefits, the operational synergy of hybrid funding comes from blending the distinct expertise and networks of its public and private partners (Block et al. 2018). This gives the startup comprehensive guidance from two critical perspectives.

- From the private partner, the venture gains essential business acumen in areas like go-to-market strategy and team building, which is known to improve its commercialization trajectory (Block et al. 2018; Hallen et al. 2014).
- From the public partner, the venture gains access to invaluable institutional resources, such as regulatory sandboxes, technology validation environments, or preferential access to public procurement contracts (Okunnuga et al. 2024; Temmes et al. 2021). Securing a government as a first customer, for instance, provides not only early revenue but also powerful market validation (Uyarra et al. 2014).

This synergistic transfer of knowledge—blending market intelligence with policy expertise—creates a uniquely capable and mature organization. A startup guided by these dual perspectives is better equipped to navigate uncertainty and scale its operations. By being both commercially savvy and institutionally connected, the venture develops a compelling profile, strongly positioning it for future growth and investment.

The Ultimate Sustainability Synergy

Addressing complex challenges like environmental sustainability requires deep collaboration, as single institutions often cannot deliver the necessary multidimensional commitments alone (Farinha 2020). Hybrid funding provides this integrated framework, where its combined structural and operational synergies create a model uniquely aligned with the dual mission of sustainable startups. In this blended arrangement, the public partner, by providing institutional access and policy expertise, validates the venture's environmental or social impact claims. Simultaneously, the private partner, by contributing commercial discipline and market-based incentives, ensures the sustainable solution is also scalable and profitable. This creates a venture where impact and profit are not competing goals but integrated components, providing the holistic support necessary for dual-mission companies to achieve lasting success.

The preceding analysis has deconstructed the investment landscape for sustainable startups into three distinct archetypes, each governed by a unique institutional logic. Private funding, guided by a market logic, is primarily optimized for financial return. In contrast, public funding operates under a state logic, prioritizing policy achievement and the correction of market failures . Hybrid funding emerges as a formal synthesis of these two, creating a blended logic that aims to leverage private-sector efficiency to achieve public interest goals. These fundamental differences in motivation translate directly into the tangible and intangible resources each archetype provides. Table 2.1 offers a comparative summary of these models, crystallizing their core differences across key dimensions. This framework, which establishes how and why these funding pathways diverge, provides the essential foundation for the development of the testable hypotheses that follow in the next section.

Table 2.1: Comparison of Funding Models

Feature	Private	Public	Hybrid
Institutional Logic	Dominated by a Market Logic.	Guided by a State Logic.	Represents a Blended Logic combining market and state elements.
Main Goal/Motivation	Primary: High financial return (realized via exit) & Secondary: Strategic benefits (for CVCs); impact objectives.	Primary: Policy achievement (innovation, regional development) & Secondary: Correcting market failures; generating positive externalities.	Primary: Mobilizing private capital for public interest goals & Secondary: Achieving synergistic outcomes (commercial viability + societal impact).
Investor Types	VC & PE funds, angel investors, corporations, impact investors.	Government agencies, development banks, government-owned VC funds, innovation funds.	Co-investment syndicates, public-private partnerships, blended-finance structures, matching funds.
Key Intangible Support			
Strategic & Operational	Hands-on guidance and active board participation focused on scaling, commercialization, and accelerating the path to a lucrative exit.	Mentorship focused on technical development, policy alignment, and long-term R&D milestones.	Blended guidance combining private-sector commercial acumen with public-sector policy insight and institutional access.
Network Access	Access to commercial networks (customers, partners, potential acquirers, downstream investors).	Access to institutional networks (government agencies, policy experts, research institutions) and preferential access to public procurement contracts.	Synergistic access to both commercial and institutional networks, bridging policy and market worlds.
Legitimacy & Validation	Provides market validation and credibility ("stamp of approval") that signals commercial potential to other market actors.	Provides a powerful 'stamp of approval' that validates the venture's technical quality for the broader market, serving as a de-risking signal for subsequent investors.	Offers dual validation: the commercial credibility of private partners combined with the institutional legitimacy of public partners.

2.4 Conceptual Framework and Hypothesis Development

The central goal of this research is to holistically compare how different funding archetypes — private, public, and hybrid — influence the success of sustainable startups. The preceding chapters have built the theoretical foundation for this inquiry, establishing that these archetypes are governed by competing institutional logics that shape their motivations and behavior. Private funding, dominated by a market logic, is primarily driven by the pursuit of financial returns; public funding, guided by a state logic, is oriented toward policy objectives and correcting market failures; and hybrid models, operating under a blended logic, emerge to synthesize these two worlds

Given these divergent motivations, applying a single, monolithic definition of "success" would be insufficient and misleading. This challenge is mirrored in academic literature, where a universally accepted definition of startup success remains elusive. Despite a vast body of research dedicated to predicting (e.g. Stuart and Abetti 1987), understanding (e.g. Kee and Rahman 2020) and measuring (e.g. Díaz-Santamaría and Bulchand-Gidumal 2021) startup success, the metrics used are highly varied, reflecting the multifaceted nature of entrepreneurial achievement. To illustrate this, an adaptation of Ben Saad (2024) overview of frequently used success dimensions in academic research is presented below.

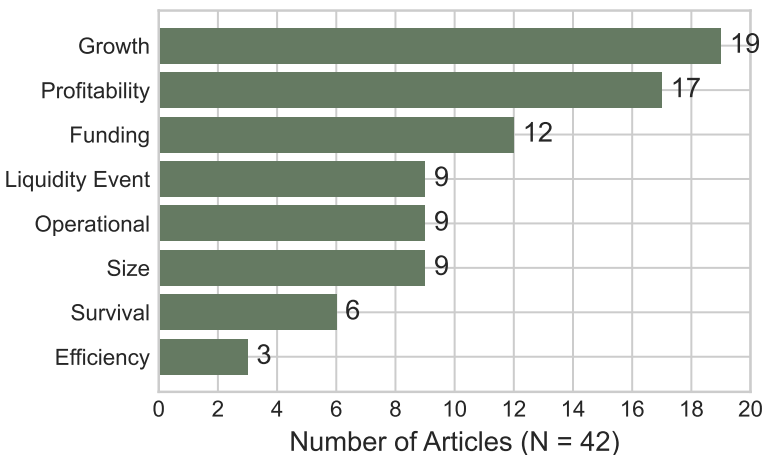


Figure 2.1: Success Metrics in Startup Literature

Recreated based on Ben Saad (2024), published in Junior Management Science under the CC BY 4.0 license.

Selected Success Metrics and Hypothesis Derivation

To account for the varied objectives of the funding archetypes, this research adopts a multi-dimensional view of success. While the literature presents numerous success dimensions, this study selects three that are well-suited to capture and measure the distinct goals of the funding archetypes: *exit*, *survival (operationality)*, and *follow-on funding*. This approach allows for a nuanced analysis where "success" is defined in terms that align with

the core logic of each funding model.

- An *Exit* (via acquisition or IPO) is the quintessential measure of success for traditional private investors, as it represents the terminal event where they realize their financial returns (Deias and Magrini 2023).
- *Survival* is a particularly relevant metric for public investors. For a startup with a long-term societal mission, remaining operational to continue its R&D and impact-generating activities is a primary indicator of success, aligning perfectly with policy goals that prioritize innovation spillovers and regional development over immediate profit. This aligns with broader economic research, where firm survival rates are often used as a key macro-economic metric to gauge economic dynamism and compare national economies (Gonzalez 2017).
- Securing *Follow-on Funding* is a fitting measure for the effectiveness of hybrid models. It signals strong market validation and investor confidence, directly testing the hypothesis that blending public de-risking with private commercial validation makes a venture more attractive to subsequent investors.

This strategic alignment of metrics to archetypes is the foundation of this study's multidimensional approach to success. However, this framework does not presume that these goals are exclusive. While an exit is the quintessential measure of success for a private fund, continued survival is an obvious prerequisite. Conversely, while public capital is patient, a successful exit can also signify mission accomplishment by launching a self-sustaining, impactful enterprise. Therefore, the hypotheses that follow are constructed to test the relative strength of each funding pathway in achieving the outcome that is most aligned with its core institutional logic.

Based on this framework, the following hypotheses are proposed:

H1 (Follow-on Funding): Sustainable startups that initially receive hybrid funding are more likely to secure subsequent rounds of follow-on funding.

The reasoning for this hypothesis is that from the perspective of a later-stage investor, an initial hybrid round makes a sustainable startup a uniquely compelling and de-risked opportunity. The presence of a public co-investor signals that the venture's technology and societal impact claims have been rigorously vetted, mitigating the high technical and policy risks characteristic of sustainable innovation. Simultaneously, the involvement of a private co-investor provides crucial market validation, confirming the startup possesses a commercially viable business model. This dual validation directly addresses the core challenge of assessing a sustainable startup's dual mission, making it an attractive candidate for follow-on funding.

H2 (Exit): The greater the prevalence of private funding in a sustainable startup's financing history, the more likely it is to achieve a successful exit, and to do so more quickly.

The reasoning for this hypothesis is that private investors, motivated by the primary goal of realizing financial returns, provide targeted non-monetary support to actively accelerate the path to an exit. For sustainable startups, this support—such as active board participation for strategic scaling and providing network access to potential acquirers—is particularly crucial for navigating long development horizons and shaping the venture into a commercially attractive acquisition target.

H3 (Survival): The greater the prevalence of public funding in a sustainable startup's financing history, the more likely it is to exhibit long-term survival.

The reasoning for this hypothesis is that public investors prioritize long-term societal impact and regional development over rapid financial returns. This patient capital approach is particularly vital for sustainable startups, which often face long R&D horizons and address market failures where their full societal value is not immediately profitable. By providing stable financing that often does not require an ownership stake, public investors foster innovation that aligns with long-term policy objectives. This insulates the venture from short-term market pressures, allowing it to focus on its core mission and thereby enhancing its prospects for long-term operational survival.

In summary, this chapter has constructed a comprehensive theoretical framework that sets the stage to analyze the success of sustainable startups under different funding conditions. It began by outlining the unique challenges and dependencies these ventures face, applying Resource Dependence Theory to explain their heightened need for external resources. Building on this, the chapter introduced Institutional Logics theory to differentiate the core motivations and behaviors of private, public, and hybrid funding archetypes. This work culminates in the development of three distinct, testable hypotheses that connect each funding archetype to a specific and aligned measure of success.

The following chapter will now bridge this theoretical foundation with empirical analysis by detailing the methodology designed to test these hypotheses. This will include a description of the data source, the operationalization of the key variables—funding type, exit, survival, and follow-on funding—and the statistical models employed to examine the proposed relationships.

3 Methodology

Building upon the theoretical framework from the preceding chapter, this section details the methodology used to empirically test the proposed hypotheses. This study adopts a quantitative, comparative, and observational research design, relying on the rigorous application of statistical models to secondary data. The following sections describe the data source and preparation process, the operationalization of key variables, the specific analytical models employed, and a discussion of the study's limitations and ethical considerations.

3.1 Data Source: *Crunchbase*

Overview and Data Sourcing

The empirical foundation for this analysis is derived from *Crunchbase*, recognized as one of the world's largest commercial databases tracking companies (Felgueiras et al. 2020). Established in 2007, *Crunchbase* offers an extensive and regularly updated repository of information pertaining to a vast array of companies (Felgueiras et al. 2020). The exact scope of its coverage is not fully defined and varies across countries and sectors. Nevertheless, its comprehensive scale makes it a valuable resource for entrepreneurial research (Den Besten 2020).

Crucially, *Crunchbase* employs several methods for collecting and validating its data. This multifaceted approach is essential for ensuring accuracy, as the database relies in part on crowd-sourced information (Crunchbase Inc. 2025c).

- *Crunchbase* Venture Program: Over 3,500 global investment firms actively contribute monthly portfolio updates in exchange for discounted access to *Crunchbase* products, ensuring a steady stream of current investment data.
- Active Community Contributors: A wide network of users submits new information and refines existing entries. These submissions undergo a rigorous process involving registration, social validation by other users, and review by human moderators.
- Artificial Intelligence: Machine learning algorithms continuously validate data accuracy, identify anomalies, and flag discrepancies for further investigation by human analysts.
- In-house Data Science Team: A dedicated team of analysts manually validates collected data, develops proprietary algorithms to enhance data quality, and performs analyses to derive business insights and reports.

Crunchbase Data Structure

The *Crunchbase* dataset is organized into distinct, interrelated tables that can be linked using unique identifiers. Broadly, the information within *Crunchbase* can be categorized into three primary types: data pertaining to (1) organizations (companies), (2) individuals (people associated with companies), and (3) investments (funding rounds and related financial activities) (Te et al. 2023). A simplified entity relationship diagram illustrating this data structure is presented in fig. 3.1.

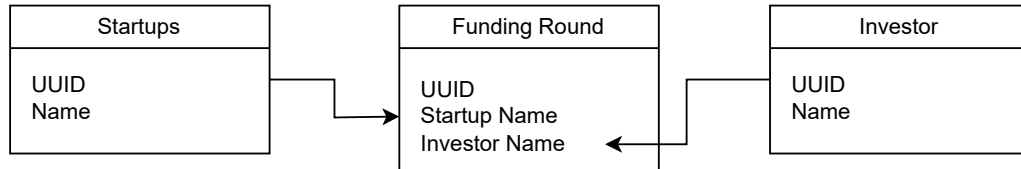


Figure 3.1: *Crunchbase* Entity Relationship Diagram

Rationale for Utilizing *Crunchbase*

The selection of *Crunchbase* as the primary data source for this study is guided by its widespread adoption in the field, its specific data strengths, and its practical accessibility for this project.

The rise of large-scale, accessible datasets has significantly contributed to the advancement of entrepreneurship research, enabling scholars to analyze thousands of companies and integrate granular information about funding events into their models (Den Besten 2020; Żbikowski and Antosiuk 2021). Among these platforms, *Crunchbase* was chosen for its superior data quality in the specific domain of this study. A comparative analysis concluded that *Crunchbase* offers commendable coverage and accuracy for financing rounds and committed capital, which is paramount for investigating funding influence on startup outcomes (Retterath and Braun 2020). Finally, on a practical level, facilitated access to the platform's professional by the TUM Entrepreneurship Research Institute version was instrumental in the execution of this research

Limitations and Potential Biases of *Crunchbase* Data

Despite the significant advantages and novel research opportunities afforded by *Crunchbase*, it is equally imperative to acknowledge its inherent limitations and potential biases. Given its hybrid sourcing strategy, the quality and comprehensiveness of the data are not solely dependent on *Crunchbase*'s internal team but also on the willingness of users and companies to frequently update their information (Te et al. 2023). This reliance on community contributions and self-reporting can introduce certain data gaps and potential biases.

Also, as highlighted by Andre Retterath (2020), the completeness of the data within *Crunchbase* may be skewed towards successful and larger companies. Specifically, larger financing rounds are often more consistently reported than smaller ones, potentially leading

to an underrepresentation of less visible or less successful ventures. These limitations necessitate careful attention during preliminary data analysis and preprocessing, including the identification and management of missing data and outliers, particularly concerning both the amount and the number of investments recorded for different companies (Ferrati and Muffatto 2020). These data-specific limitations are considered throughout the analytical process and are further discussed in the broader limitations section (Section 3.6) of this chapter.

3.2 Data Access and Inclusion Criteria

The empirical analysis began with the acquisition of the raw dataset from Crunchbase, obtained via a professional (Pro) license subscription. Data was extracted as CSV exports of the core entity tables — including organizations, funding rounds, and investors — to establish a comprehensive data pool. This original, unfiltered dataset served as the foundation to which the following inclusion criteria were applied.

The initial phase of data preparation involved applying specific inclusion criteria to the raw dataset to serve as the basis for all subsequent cleaning, preprocessing, and analysis. The guiding principle at this stage was to impose minimal initial restrictions, thereby ensuring a broad and comprehensive data pool from which to draw insights. The specific criteria applied to achieve this are detailed below.

Founding Date

To ensure that the included startups were both sufficiently mature for observable outcomes (such as exits, follow-on funding, or cessation of operations) and relevant within a contemporary entrepreneurial context, the dataset was constrained to companies founded within the period of 2010 to 2020. This specific window was selected based on the following considerations:

- *Lower Bound (2010)*: While *Crunchbase* itself was established in 2007 (Crunchbase Inc. 2025a), data preceding 2010 tends to be less comprehensive and potentially inconsistent. Beginning the observation period in 2010 mitigates issues related to early database patchiness and enhances data reliability.
- *Upper Bound (2020)*: This cut-off point allows for a meaningful observation period for startup lifecycles. Although startup exit timelines can vary considerably, some sources suggest an average exit timeframe of approximately five years (WilmerHale 2024). Extending the founding window to 2020 ensures a sufficiently broad timeframe, thereby yielding an ample number of data points for robust statistical analysis while still allowing for the observation of subsequent funding and exit activities.

Funding Activity

Given that funding dynamics constitute the central focus of this analysis, it was imperative to ensure that all observed startups had, in fact, secured funding. Consequently,

only startups explicitly recorded with at least one "Investment round" were retained in the dataset. From a temporal perspective, all funding activity for these startups was tracked and observed through the end of 2024, providing a comprehensive view of their capital trajectories and enabling the identification of long-term patterns related to funding acquisition and success metrics.

Industry Focus: Sustainable Startups

This research specifically targets sustainable startups. To identify this sample, the study deliberately utilized *Crunchbase's* predefined "Sustainability" industry group. While more sophisticated classification, such as using Natural Language Processing to analyze company descriptions have been employed in academic literature (e.g. Batista and Carvalho 2015; Felgueiras et al. 2020), the choice to use *Crunchbase's* established labels was made to ensure methodological consistency and replicability, while avoiding the subjective bias inherent in manual classification schemes. Given that the concept of "sustainability" is itself inherently broad and dynamic, aligning with *Crunchbase's* established logic provides a transparent and robust foundation for the analysis. This "Sustainability" group is a comprehensive category, encompassing 23 distinct industries that reflect a wide spectrum of sustainability-focused activities:

Biofuel, Biomass Energy, Carbon Capture, Clean Energy, CleanTech, Energy Efficiency, Environmental Engineering, Geothermal Energy, Green Building, Green Consumer Goods, GreenTech, Hydroelectric, Natural Resources, Organic, Pollution Control, Recycling, Renewable Energy, Solar, Sustainability, Waste Management, Water Purification, Wildlife Conservation, Wind Energy

A complete list of these industries, including their descriptions as provided by *Crunchbase*, can be found in Appendix A.1.

3.3 Data Cleaning and Preprocessing

The data acquisition process yielded three primary interconnected entity tables — Funding Rounds, Startups, and Investors - each linked by unique identifiers (as illustrated in Figure 3.1). This initially rich dataset comprised 19465 funding rounds, 8023 startups, and 11869 investors. However, consistent with expectations for large-scale secondary data, this raw data was characterized by inherent incompleteness, inconsistencies, and the presence of outliers. Consequently, a rigorous and iterative data cleaning process was applied to enhance data quality and ensure the robustness of the subsequent analyses.

Consistency and Logic Checks for Funding Rounds

The initial phase of data cleaning commenced with an assessment of the funding rounds data, focusing on consistency and logical coherence across various attributes. Specifically, the dates of individual funding rounds were cross-referenced against the corresponding startup's founding date and any recorded exit dates:

- *Founding Date Consistency*: Funding rounds dated prior to a startup's official founding date were identified and removed. This step ensured the chronological integrity of the data, as a funding event logically cannot precede the establishment of the company itself.
- *Post-Exit Funding Removal*: Similarly, funding rounds occurring after a startup's recorded exit date were omitted. This filtration aimed to maintain the analytical focus on growth and operational financing, excluding any potential post-acquisition or post-IPO financial events that fall outside the scope of early-stage venture dynamics.

Completeness Checks and Startup Omission Strategy

Following the initial consistency checks, a thorough completeness assessment was performed on critical attributes of the funding events. Instead of merely removing individual inconsistent funding rounds, a more stringent approach was adopted: if any of the specified crucial information was missing for a funding round, the entire startup and all its associated funding events were omitted from the dataset. This comprehensive removal strategy was implemented to prevent the introduction of bias that could arise from selectively retaining startups with incomplete funding histories, thereby ensuring a higher degree of data integrity for the remaining observations.

The attributes checked for completeness in each **funding round** included:

- *Investor Name and URL*: The presence of both an investor's name and a unique URL was mandated. This ensured sufficient identifying information to consistently and accurately classify investor types (e.g., public, private, or hybrid), a central variable in this research.
- *Funding Amount per Round*: The monetary value of each funding round was required. This allowed for the inclusion of a financial dimension in the analysis and served as a proxy for data quality, operating on the premise that funding rounds with explicitly listed amounts generally exhibit a higher degree of accuracy and formalization.
- *Crowdfunding Rounds*: Funding rounds identified as crowdfunding events were systematically excluded. This exclusion was driven by the research's focus on institutional investor behavior, as crowdfunding mechanisms typically involve a different investor base and distinct motivations that are outside the scope of this study.

Enhancing Dataset Robustness and Relevance

Subsequent to the completeness checks, additional logical criteria were applied to refine the dataset further, aiming to enhance its robustness and ensure that the retained startups were most relevant to the study's core objective of comparing different institutional investor approaches. The primary goal was to focus on startups demonstrating a certain level of investor maturity and institutional backing, while excluding those primarily funded by

individual investors or through highly syndicated, potentially atypical, rounds.

This refinement involved the following exclusion logic, which was applied at the startup level based on aggregated funding characteristics:

- *Removal of Startups with Predominantly Inexperienced Investors:* Startups where the ratio of investments from 'inexperienced' investors (defined as investors with 10 or fewer prior recorded investments) to their total investors exceeded 70% were excluded. This aimed to filter out ventures primarily reliant on less experienced capital, allowing for a clearer focus on the influence of more seasoned investor types
- *Exclusion of Angel-Dominated Startups:* Startups where angel investors constituted more than 50% of their total investors were removed. This measure ensured that the dataset's focus remained on companies attracting institutional or formalized venture capital, rather than those predominantly supported by individual angel funding.
- *Exclusion of Highly Syndicated Rounds:* Startups that had any single funding round with more than 10 participating investors (a high syndicate size) were excluded. Such highly syndicated rounds can sometimes indicate unique funding structures or atypical investor dynamics that may not align with the general patterns of institutional investment under investigation.

Again, these criteria were applied disjunctively, meaning a startup was removed if it met any of the defined conditions, thereby ensuring a dataset more representative of ventures with a significant degree of institutional investor engagement.

Finally, to ensure the geographical context essential for complete control variable construction, any remaining startups that lacked a designated headquarters (HQ) location were also removed from the dataset.

3.4 Research Design and Feature Engineering

Following the data cleaning and preparation steps, the next phase of the methodology involved significant feature engineering. To test the hypotheses developed in the theoretical chapter, it was necessary to transform the event-level data from the interconnected funding and investor tables into a single analytical dataset. The objective of this process was to construct the core dependent and independent variables where the startup serves as the primary unit of analysis for all subsequent modeling.

3.4.1 Dependent Variables

As established in the theoretical framework, this study measures success across multiple dimensions to align with the distinct goals of each funding archetype. Accordingly, the following three dependent variables were constructed to capture these different facets of a sustainable startup's outcome.

Follow-On Funding

To investigate the influence of the initial investor type on a startup's subsequent growth trajectory, the concept of "follow-on funding" was operationalized using two distinct variables, capturing both the occurrence and the financial scale of the event.

- First, to measure the likelihood of attracting further investment, the binary variable **Follow-On Funding** (`bin_follow_on_funding`) was constructed. Coded as '1' if a startup received any funding after its initial round and '0' otherwise, it serves as an indicator of early perceived success and continued investor confidence.
- Second, to capture the financial magnitude of this success, the variable **Logarithmic Follow-On Amount** (`log_follow_on_amount`) was created. This was derived by taking the total U.S. dollar amount of all follow-on funding, converting it to millions, and applying a log-plus-one transformation ($y = \log(x+1)$) to normalize the skewed data and accommodate cases with zero follow-on funding.

Exit

Hypothesis H2 investigates the influence of funding on both the likelihood and speed of a startup exit. To test this dual-natured hypothesis, two dependent variables were constructed.

- First, to measure the likelihood of an exit, the binary variable **Exit Status** (`bin_exit_status`) was created. Consistent with established metrics in the literature, an exit is defined as either an acquisition or a successful IPO (Potanin et al. 2023). Accordingly, the variable is coded as '1' (Yes) if a startup experienced such an event and '0' (No) otherwise. For companies with multiple liquidity events, the date of the chronologically earlier event was used as the definitive exit date.
- Second, to analyze the speed of an exit using survival analysis, the variable **Time to Exit** (`time_to_exit_years`) was engineered. This variable measures the duration in years from a startup's founding date to an outcome. For startups that exited, this is the time until their recorded exit date. For those that had not exited by the end of the observation period (April 20, 2025), the duration is the time until this cut-off date. This approach correctly handles the right-censored data required for time-to-event modeling.

Together, these variables enable a comprehensive test of a startup's path to a liquidity event.

Operational Status (Survival)

The final hypothesis concerns firm survival, which was operationalized as the binary variable **Operating Status 2025** (`bin_op_status_2025`). The construction of this variable required a custom approach to overcome two significant challenges in the source data.

The first challenge was the fragmented and often missing closure dates for non-operational

startups in *Crunchbase*. The lack of reliable event dates made a time-to-event survival analysis (e.g., a Cox model) unfeasible. Consequently, the study was methodologically constrained to assess operational status at a fixed point in time (May 2025) using logistic regression. This necessary trade-off introduces limitations, particularly regarding time-dependent biases, which are addressed in section 3.6.

The second challenge was the reliability of *Crunchbase*'s native "Operating Status" label itself. Initial robustness checks revealed that many startups listed as "active" were, in fact, defunct. Therefore, a manual double-verification process was implemented to create a more accurate classification. This involved assessing each startup's webpage activity, an approach supported by research that considers an inactive homepage a strong indicator of market failure (Gautam and Wattanapongsakorn 2024). In ambiguous cases, professional networking sites and public company registries were also consulted.

The resulting variable, indicates a startup's verified operational status, coded as '1' for operational and '0' for inactive or closed.

3.4.2 Independent Variables

The explanatory power of the analytical models hinges on the independent variables constructed for this study. To test the core hypotheses and isolate their effects, these variables were separated into two categories: the primary independent variables, which operationalize the funding archetypes, and a comprehensive set of control variables that account for other known influences on startup success. The construction of both variable types, drawing upon *Crunchbase*'s rich data and tailored feature engineering, is detailed below.

Funding Type Classification

A central component of this study was to classify each funding round as 'public', 'private', or 'hybrid'. This round-level classification is essential for constructing the primary independent variables that measure the influence of each funding archetype on a startup. To achieve this in a systematic and replicable manner, a two-step process was employed:

1. Each individual investor was classified into one of the three archetypes.
2. This investor-level data was then aggregated to assign a final type to each funding round.

Step 1: Classifying Individual Investors

The classification process began at the individual investor level, a task that presented two primary challenges. The first, as outlined in the theoretical chapter, pertains to the inherent fluidity in the boundaries between investor types. This is compounded by the often incomplete disclosure of an investor's precise 'composition' or funding sources.

To ensure a rigorous and consistent classification despite these challenges, the following rules were applied:

- *Public Investors*: Classified as such if unequivocally owned and governed by public bodies. This category includes national and international agencies, nationally owned development banks, and entirely government-funded VC funds.
- *Private Investors*: Assigned to this category if no discernible public involvement was apparent. This encompasses conventional VC funds, PE firms, CVC, and individual or angel investor groups.
- *Hybrid Investors*: Designated as hybrid if any indication of public-private interplay was present. This primarily includes venture capitalists with governmental backing, university-affiliated programs, or organizations structured as public-private partnerships.

While the aforementioned logic guided the individual investor classification, the methodology for achieving this required a multi-source approach. The native *Crunchbase* "Investor Type" label provided an initial indication of an investor's nature; however, its predefined labels were deemed overly broad and, upon a robustness check, demonstrated insufficient accuracy for the granularity required by this study. Therefore, additional sources were consulted: (1) the expanded *Crunchbase* profile, specifically its 'About' section; (2) the investor's LinkedIn profile, focusing on their brief company description; and (3) the investor's official website, with particular attention to 'About Us' and 'Partners' sections, as well as website headers or footers often indicating affiliations. This largely manual classification process aligns with approaches adopted in prior research, such as Brander, Du, et al. (2015), who conducted extensive website analysis of VC firms to identify indicators of government support. This method, however, relies on the critical assumption that investors transparently disclose their public affiliations, particularly in cases of co-funding or when public institutions serve as Limited Partners. While this assumption is supported by the general benefits of voluntary disclosure in enhancing credibility (Shehata 2014) and the strategic advantages associated with public affiliations, the possibility of some misclassifications due to incomplete or non-transparent disclosure cannot be entirely eliminated. This inherent limitation will be further discussed in Section 3.6

Step 2: Assigning a Type to Funding Rounds

Following the comprehensive classification of individual investors, their respective types were systematically assigned to the funding rounds they participated in. This presented a final challenge in the context of syndicated rounds involving multiple investor types. For rounds with a single investor or a homogeneous group, the classification was straightforward. However, for mixed-syndicate rounds, a nuanced "dominance logic" was applied.

A specific logic was introduced to classify such mixed funding rounds as 'hybrid', even if a 'hybrid' investor was not explicitly present in the syndicate. A funding round was deemed 'hybrid' if it featured a significant blend of public, private, or classified hybrid investors. To avoid an over-representation of 'hybrid' rounds and maintain a 'dominance logic', a conservative threshold of 0.67 (two-thirds) was established. This meant if a funding round

featured more than two-thirds of investors belonging to a single type (public, private, or hybrid), the round retained that specific type; otherwise, it was classified as a 'hybrid' round. While this specific logic is novel in the existing literature, it was deemed necessary to enhance the robustness of the dataset and ensure that the classification reflected a clear dominant type or a deliberate mixed composition. This sensitive approach was favored over simply classifying any mixed appearance as a 'hybrid' round, which might have been overly susceptible to errors arising from the initial, somewhat manual, investor labeling process.

The determined classification for each funding round was then utilized to construct several startup-level variables. To capture the "prevalence" of each funding archetype in different ways, three sets of metrics were engineered:

- **Count:** The variables `num_public_rounds`, `num_private_rounds`, and `num_hybrid_rounds` represent the simple cumulative count of each funding round type throughout a startup's lifecycle.
- **Proportion:** To measure relative influence, ratio variables (e.g., `ratio_private_rounds`) were created by dividing the count of each round type by a startup's total number of funding rounds.
- **Financial Magnitude:** To capture the financial contribution of each archetype, a third set of variables (e.g., `amount_private_funding_m`) was calculated by summing the funding amount from all rounds of a given type, measured in millions of U.S. dollars.

Furthermore, the classified type of a startup's first and last tracked funding rounds (`first_round_funding_type` and `last_round_funding_type`) were assigned to the startup's profile, reflecting the investor composition at these critical junctures.

Moderating and Control Variables

In addition to the primary independent variables that operationalize the funding archetypes, two further sets of variables were constructed. The first is a key moderating variable used to explore the structural characteristics of investments. The second is a comprehensive set of control variables included in the analytical models to account for confounding factors and ensure the robustness of the findings.

Moderating Variable: Funding Nature

To gain deeper insights into the structural characteristics of investments, the Funding Nature was constructed as a binary variable (dilutive vs. non-dilutive). This was created by mapping *Crunchbase's* native funding types (Note: the platforms use of the terminology type is different to this thesis) to the appropriate classification (the specific logic is detailed in Appendix A.2). Given its strong theoretical relationship with the primary funding archetypes, this variable was specifically used as a moderator to test whether the influence of a funding type (e.g., public) on startup success differs based on whether it was

provided as equity or not.

Control Variables

The following control variables were included in the analytical models based on academic standards and data availability to account for other factors that could influence startup outcomes:

- *Startup Maturity*: As a firm's age can independently influence its success, startup age (e.g. `age_at_exit`, `age_at_first_round`) was calculated as the time elapsed between founding and a specific event date.
- *Startup Size*: A startup's size was measured by its total funding amount. This was included as a control variable because it is a key determinant of a venture's capacity to scale and absorb shocks. Depending on the analysis, this metric was used in two ways: either as the cumulative lifetime funding or as the amount from a specific round (e.g., the first or last).
- *Macroeconomic Context*: The date of various events (e.g., funding rounds, founding, exit) was used to implicitly account for broader temporal trends and economic conditions.
- *Geographic Region*: The geographical region of a startup, derived from its HQ location (following an approach similar to Gautam and Wattanapongsakorn (2024), was merged into broader clusters. This clustering adapted, but did not strictly adhere to, the logic proposed by the United States Geoscheme (Divisio 2025) and the World by Region classifications (Group 2017).
- *Startup Industry Cluster*: To address *Crunchbase's* often loose and inconsistent industry labels (which can appear as comma-separated lists), individual industries were grouped into higher-level clusters. This process created a more robust categorical variable for use in the analytical models.

A comprehensive list of all independent variables, including their brief definitions, is provided in Appendix A.3. All categorical independent variables were subsequently converted into dummy (binary) variables through one-hot encoding for inclusion in the statistical models. It should be noted that the specific set of variables used was tailored for each of the three analytical models, as detailed in the upcoming section.

3.5 Analytical Models

3.5.1 Analytical Model for H1: Follow-on Funding

To test Hypothesis 1, a multi-stage analytical approach was employed to investigate the influence of the initial investor type on a startup's ability to secure subsequent financing.

The primary analysis utilized a binary logistic regression model to predict the likelihood of receiving any follow-on funding using the binary follow-on status variable. The key

independent variable was the operationalized first round funding type. To isolate its effect, the model controlled for several factors, including: the size of the initial investment, the startup's age at the time of the round, macroeconomic timing (via the funding year), and fixed effects for geographic region and industry cluster.

To delve deeper into the nuances of this relationship, two post-hoc analyses were conducted:

1. **Linear Regression on Funding Amount:** To assess whether the initial investor type also influenced the financial magnitude of subsequent investment, a separate Ordinary Least Squares (OLS) regression was performed using the logarithmic follow-on amount as the continuous dependent variable.
2. **Interaction Effect Analysis:** To explore the interplay between investor type and investment structure, the primary logistic regression model was extended to include an interaction term. This tested whether the effect of the first round investor type was moderated by the first round funding nature (i.e., dilutive vs. non-dilutive).

For all analyses, a consistent procedure was followed: a baseline model containing only control variables was compared to the full model using a Likelihood Ratio Test (LRT) to confirm the statistical significance of the added variables. Furthermore, a Variance Inflation Factor (VIF) check was conducted prior to final estimations to ensure the stability and reliability of the model coefficients (All VIF results can be found in Appendix A.4, for the follow-on model see Table A.4).

3.5.2 Analytical Model for H2: Likelihood and Speed of Exit

Hypothesis 2 posits that a greater prevalence of private funding increases both the likelihood and speed of a successful exit. Before detailing the specific models, three key methodological points should be noted.

First, a preliminary analysis of exit events by startup founding year was conducted. As shown in Figure 3.2, this did not reveal a clear time-correlated trend, why the decision was made to retain the full dataset for analysis rather than applying a temporal cut-off. Second, to best operationalize the "prevalence" of funding, several model specifications were tested using different independent variables (round counts, financial magnitude, and proportional ratios). The final models presented are those that yielded the best fit, as determined by the lowest Akaike Information Criterion (AIC) — a statistical measure that estimates prediction error, thereby balancing model fit with model complexity. Third, while using a startup's entire funding history is an aggregate measure, this approach is consistent with prior academic work, which has drawn conclusions from even more streamlined data, such as the appearance of a single funding round (Brander, Du, et al. 2015).

Given the dual nature of the hypothesis, a two-part analytical strategy was employed.

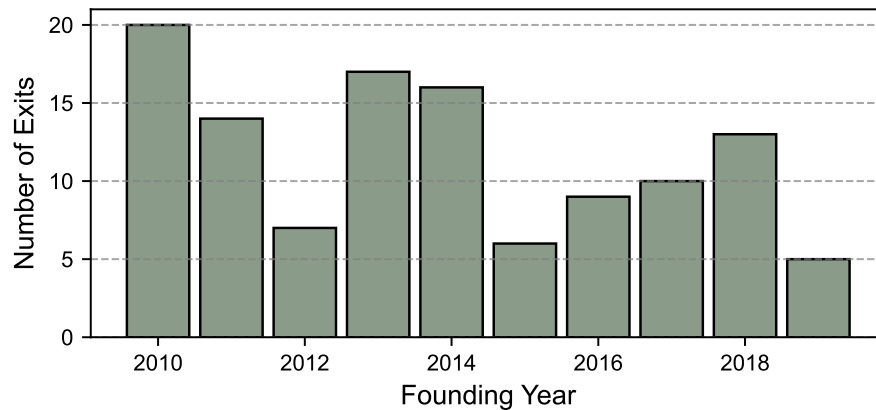


Figure 3.2: Distribution of Exited Startups by Founding Year

Part 1: Likelihood of Exit

To test the likelihood of an exit, a binary logistic regression model was used, with the Binary Exit Status as the dichotomous dependent variable. The "prevalence" of funding types was operationalized using the proportional ratio variables, as this specification achieved the best AIC score. To isolate the effect of these variables, the model controlled for: total funding amount, the startup's founding year, its age at the last funding round, the total number of funding rounds, the amount of the last funding round, as well as fixed effects for geographic region and industry cluster.

Part 2: Speed of Exit

To test the speed of an exit, a Cox Proportional Hazards (CoxPH) model was employed, which is well-suited for time-to-event data. The model used the previously described Time to Exit as the duration variable and the binary exit status as the event indicator. The same set of independent (ratio variables) and control variables from the logistic regression model were used as covariates.

Again, For both models, diagnostic checks for multicollinearity were performed using the (see Table A.5). Furthermore, another confirmed that including the funding ratio variables provided a statistically significant improvement over a baseline model containing only controls.

3.5.3 Analytical Model for H3: Long-Term Survival

The final hypothesis, focusing on long-term survival, required a specific analytical strategy to navigate significant data limitations. As detailed previously, the lack of reliable closure dates in the source data made a time-to-event survival analysis unfeasible. This constrained the methodology to a cross-sectional analysis at a fixed point in time.

Furthermore, a preliminary analysis revealed a strong correlation between a startup's age and its likelihood of being closed (as illustrated in Figure 3.3). To control for this and specifically analyze long-term endurance, the sample was filtered to include only mature

startups with a minimum age of 10 years. While this isolates the desired phenomenon, it necessarily introduces a survivor bias, which is addressed in the limitations section (Section 3.6).

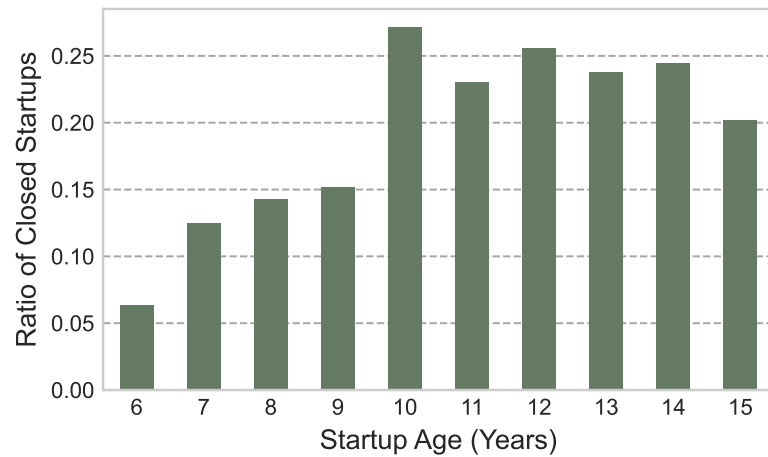


Figure 3.3: Ratio of Closed Startups by Startup Age in Years

To test the influence of funding composition on the survival of these mature firms, a nested binary logistic regression analysis was conducted. The dependent variable was the binary operating status in 2025. The analysis involved comparing three sequential models to systematically determine which level of funding detail held the most explanatory power:

1. **Model 1 (Baseline):** This model included only control variables such as the startup's founding year, industry, region and total number of funding rounds.
2. **Model 2 (Archetype Amount):** To the baseline, this model added variables representing the total financial magnitude of funding received from each of the three archetypes (public, private, and hybrid).
3. **Model 3 (Archetype & Nature Amount):** The final model further disaggregated the financial variables by both their archetype and their nature (e.g., private-dilutive, public-non-dilutive, etc.).

This nested approach allowed for a rigorous test of whether the source and, ultimately, the structure of capital provide progressively more explanatory power. The fit of each model was once more formally compared using a LRT, and a VIF check was conducted to ensure the absence of problematic multicollinearity (see Table A.6).

All data preparation, statistical modeling, and visualization for this study were performed using the Python programming language. The analysis relied on its ecosystem of established scientific libraries, primarily including `pandas` for data manipulation, `statsmodels` for regression analyses, and `matplotlib` and `seaborn` for data visualization.

3.6 Methodological Limitations

Already touched upon repeatedly throughout this chapter, the primary limitations stem from the nature of the secondary data source, *Crunchbase*. As discussed in section 3.1, while *Crunchbase* is a comprehensive resource, it has known biases. Its coverage tends to favor more successful, venture-backed, and U.S.-based technology firms. Although some of this study's filtering processes mitigate some of these issues, a potential bias toward more visible startups may remain.

Beyond these known sampling biases, the analysis fundamentally assumes that the underlying event data is both complete and accurate. The validity of the findings is contingent on *Crunchbase* having comprehensively tracked the full funding history and correctly reported the exit status for each startup in the sample. Any unobserved missing funding rounds or inaccurately recorded exit data represents a critical limitation that could influence the results.

Furthermore, the operationalization of key variables presented specific challenges:

- *Investor Type Classification*: The classification of investors into public, private, and hybrid archetypes, while based on a systematic multi-source verification process, relied on publicly available information. The potential for incomplete disclosure by investors regarding their Limited Partners or governance structures means that some misclassification is unavoidable. The difficulty was compounded by the need to interpret websites in various languages and to consolidate the nature of an investment entity when multiple individual investors were involved in a single round.
- *Operational Status and Survivor Bias*: As detailed above, the significant number of missing closure dates for failed startups made a traditional time-to-event analysis for survival unfeasible. The decision to instead analyze long-term operationality by filtering the sample to firms aged 10 years or older was a necessary methodological trade-off. However, this introduces a significant survivor bias for the analysis of H3. The findings on survival are therefore only applicable to the population of firms that have already survived for a decade and cannot be generalized to predict the survival probability of a startup from its inception.

Finally, the diagnostic checks for multicollinearity revealed a moderate level of collinearity among the regional control variables. The VIF scores for some regions approached 6.5, a likely result of the high concentration of startups in North America and Western Europe. As these values remain below the common critical threshold of 10, the main model results are considered robust. Nevertheless, this moderate collinearity is acknowledged as a limitation that could slightly affect the precision of the coefficients for the regional variables themselves.

Causality and Unobserved Factors

Beyond the data and measurement challenges discussed above, a more fundamental limitation arises from the study's observational research design, which means the findings represent strong associations rather than strict causal relationships. While the models include a comprehensive set of controls, it is impossible to account for all unobserved heterogeneity that could influence a startup's trajectory.

For instance, factors such as the intrinsic skills and network of the founding team, or the specific competitive intensity within an industry sub-sector, are notoriously difficult to measure yet are known to influence success (Żbikowski and Antosiuk 2021). These unobserved variables could create an omitted variable bias; a highly skilled founding team, for example, might be more likely to attract private funding and be more likely to achieve an exit, independent of the funding's direct effect. Consequently, the results should be interpreted as demonstrating the powerful and distinct pathways associated with each funding archetype, rather than as definitive proof of a causal link.

3.7 Ethical Considerations

Lastly, the research presented in this thesis was conducted with strict adherence to ethical principles and academic integrity. The study relies exclusively on available secondary data sourced from *Crunchbase* and publicly accessible web data for enrichment. All data pertains to organizations and their institutional funding activities; no sensitive personal data of individuals was collected, used, or stored during the research process. The methods of data collection and analysis were designed to be objective and transparent, ensuring the replicability and reliability of the findings.

4 Findings

This chapter presents the empirical results of the statistical analyses designed to test the three hypotheses developed in the theory chapter. The section begins with a descriptive overview of the final analytical sample, followed by a sequential presentation of the findings for each hypothesis: **H1 (Follow-on Funding)**, **H2 (Exit)**, and **H3 (Survival)**. The results are presented objectively, with interpretation and theoretical implications reserved for the Discussion chapter.

4.1 Descriptive Statistics

This section provides a descriptive overview of the final dataset to facilitate a comprehensive understanding of the data that informs the subsequent inferential models. The summary is structured around the three core entities of the dataset—startups, funding rounds, and investors—and focuses primarily on the characteristics of the independent and control variables. Descriptive statistics for the dependent variables are reserved for their respective analytical sections to provide immediate context for the hypothesis tests. After the data cleaning and filtering process detailed in the previous chapter, the final sample consists of 1,638 unique sustainable startups, which collectively engaged in 2,809 funding rounds from 2,716 distinct investors between 2010 and 2024.

4.1.1 Startups

The startups forming the basis of this analysis represent a diverse landscape of sustainability-oriented innovation. Methodologically, the sample is unified by its classification within *Crunchbase's* "Sustainability" industry group, a broad category covering 23 distinct industries to ensure a focus on ventures with an environmental or social mission. To illustrate the specific focus areas within this cohort, Figure 4.1 presents a word cloud of the most frequent industry labels. The resulting visualization, with its strong emphasis on terms like "Renewable Energy," "Energy," "CleanTech," and "Waste Management," confirms that the dataset is heavily populated by ventures at the forefront of the green transition.



Figure 4.1: Distribution of Top Startup Industries

To use industry as a control variable, the 391 unique industry labels were grouped into broader clusters. As shown in table 4.1, the results confirm the dominance of energy-related ventures, with *Sustainable Energy & Power* constituting the largest single cluster. This is followed by *Environmental & Circular Economy*, indicating a significant presence of firms focused on waste management and recycling solutions.

Table 4.1: Distribution of Startups by Industry Cluster

Startup industry cluster	Count
Sustainable Energy & Power	613
Environmental & Circular Economy	369
General Sustainability / Other	206
Sustainable Manufacturing & Industrial	195
Sustainable Digital & AI Solutions	103
Sustainable Ag & Food Systems	99
Sustainable Mobility & Transportation	53

The geographical distribution of these startups, visualized in the world map in fig. 4.2 reveals a significant concentration in North America and Western Europe. While sustainable innovation is a global phenomenon, the data suggests that the most visible and well-funded ventures are predominantly located in these two regions, a finding likely influenced by both the maturity of their venture capital markets and *Crunchbase*'s inherent data collection biases.

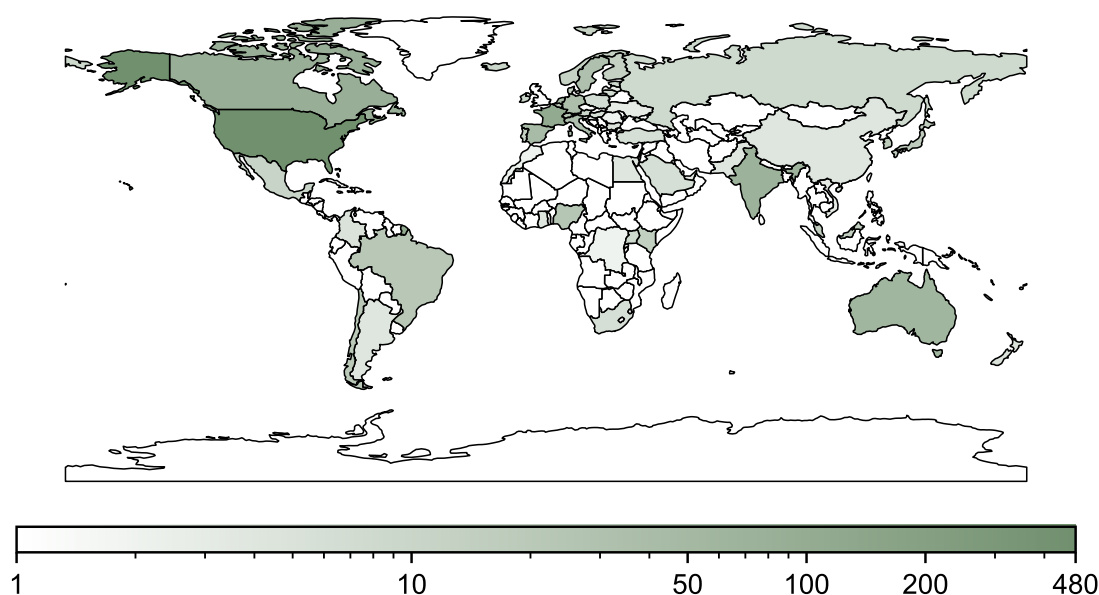


Figure 4.2: Distribution of Startup Headquarter Locations

This concentration is further detailed in table 4.2, which shows the precise counts per regional cluster. North America and Western Europe together account for over 70% of the startups in the sample. This clustering is crucial for the subsequent analysis, as it allows for controlling for broad regional differences in policy, economic conditions, and investment culture.

Table 4.2: Distribution of Startups by Clustered Region

Region	Count
Western Europe	589
North America	578
Asia Pacific	172
South Asia & MENA	119
Africa	68
Lat. America	57
Central & Eastern Europe	55

A numerical profile of the startups, presented in Table 4.3, reveals several key characteristics of the sample. The ventures have a mean age of approximately 10 years, a figure that is a direct result of the sampling window for firms founded between 2010 and 2020. The funding data displays highly skewed distributions for both the capital raised and the number of investment rounds—a classic feature of venture ecosystems. While the mean total funding is a substantial \$46.4 million, the median is a more modest \$1.64 million, indicating the average is heavily influenced by a small number of 'mega-funded'

outliers. This pattern is mirrored in the financing frequency: the mean number of funding rounds per startup is 1.71, while the median is just one. Taken together, these metrics paint a picture of a landscape where the typical startup secures a single round of institutional funding, while a smaller cohort progresses to raise significant capital over multiple subsequent rounds.

Table 4.3: Descriptive Statistics for Startups in the Sample

	Mean	Std. Dev.	Min	Median	Max
Startup age (years)	10	3	6	10	15
Total funding (\$ millions)	46.4	274.8	0.002	1.64	8,946
No. of funding rounds	1.71	1.38	1	1	13

4.1.2 Funding Rounds

Having profiled the startups, the analysis now turns to the characteristics of the funding events themselves. The 2,809 distinct funding rounds in the dataset show clear patterns that align with the theoretical archetypes discussed previously.

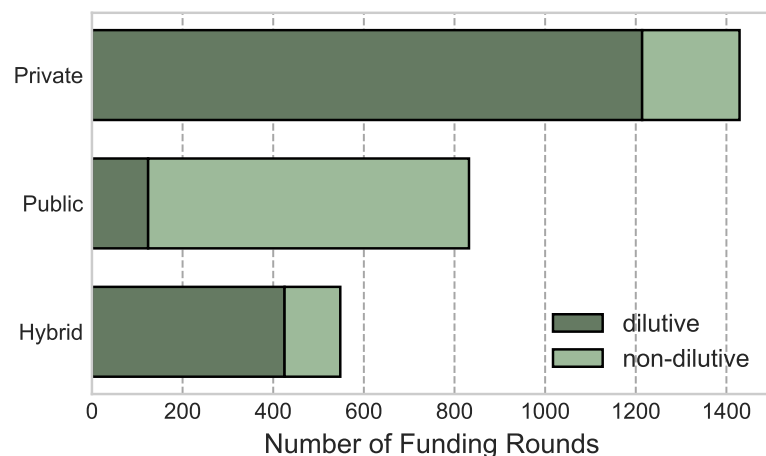


Figure 4.3: Distribution of Funding Rounds by Type and Nature

A foundational observation is the strong relationship between a funding round's archetype and its financial nature—that is, whether it is dilutive (equity) or non-dilutive (e.g., grants, loans). Figure 4.3 illustrates this relationship. The vast majority of private funding rounds are dilutive, reflecting the market logic's focus on acquiring ownership stakes. Conversely, public funding rounds are overwhelmingly non-dilutive, consistent with the state logic's aim to provide support without taking equity. Hybrid rounds, as expected, exhibit a slightly more balanced composition, blending both dilutive and non-dilutive structures, but tend towards the adaptation of dilutive investments as well.

Table 4.4 provides the precise numerical breakdown that underlies fig. 4.3, detailing the absolute counts and financial volumes for each combination of type and nature.

Table 4.4: Funding Round Statistics by Type and Nature

Funding Type	Nature	Rounds	Median Amount (USD)	Mean Amount (USD)
Private	Dilutive	1,214	2,933,723	34,601,841
	Non-dilutive	215	450,000	85,505,473
Public	Dilutive	124	328,877	12,241,649
	Non-dilutive	708	249,958	8,049,630
Hybrid	Dilutive	425	1,190,140	8,809,137
	Non-dilutive	123	138,993	38,251,597

Beyond the nature of the rounds, their financial scale also differs significantly across the archetypes, a trend that is already apparent from the median values in table 4.4. Figure 4.4 visualizes this divergence in greater detail, using box plots to compare the distribution of funding amounts for each type. The plot, which uses a logarithmic scale to account for the wide range of values, confirms that private rounds have the highest median funding amount, indicating a typical private investment is larger than its public or hybrid counterpart. However, the visualization also underscores a feature common to all three archetypes: the presence of numerous high-value outliers, highlighting the “power-law” dynamic of venture finance where a few deals are orders of magnitude larger than the rest.

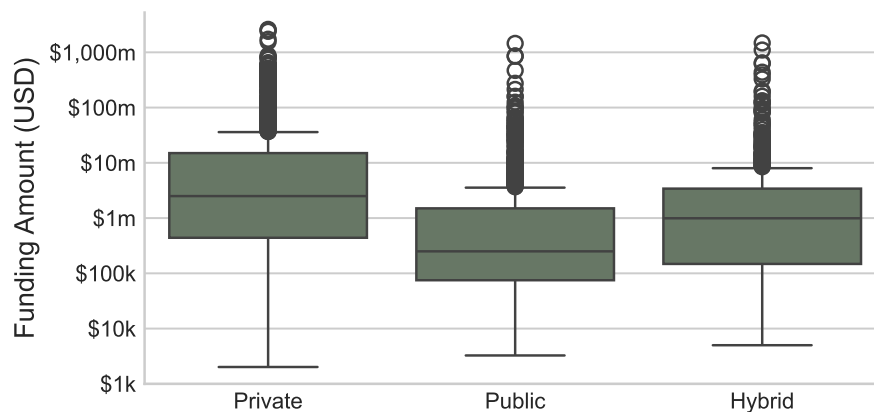


Figure 4.4: Distribution of Funding Amount across Funding Type

Table 4.8 provides a detailed summary of the 1,429 private, 832 public, and 548 hybrid funding rounds identified in the sample. The statistics reinforce the visual evidence from the box plot regarding financial scale; the median funding amount for a private round (\$2.5 million) is ten times larger than for a public round (\$250,000). The table also highlights distinct syndication behaviors: private and hybrid rounds tend to involve more investors (a mean of 2.19 and 2.01, respectively), whereas public rounds are typically sourced from a single entity.

Table 4.5: Descriptive Statistics of Funding Rounds by Archetype

Statistic	Private	Public	Hybrid
Funding Amount (USD)			
Mean	42,260,539	8,674,401	15,417,572
Median	2,500,000	250,000	992,048
Std. Dev.	161,366,072	69,517,402	93,417,238
Number of Investors per Round			
Mean	2.19	1.03	2.01
Median	1.00	1.00	2.00
Total Rounds (N)	1,429	832	548

Finally, the evolution of funding activity over time is shown in fig. 4.5. The stacked area chart illustrates a clear and substantial increase in the total number of funding rounds for sustainable startups from 2010 to a peak around 2022. The chart also reveals a changing composition over time, with private funding consistently making up the largest share of activity, but with public and hybrid funding playing an increasingly significant role, particularly in the later years of the observation period.

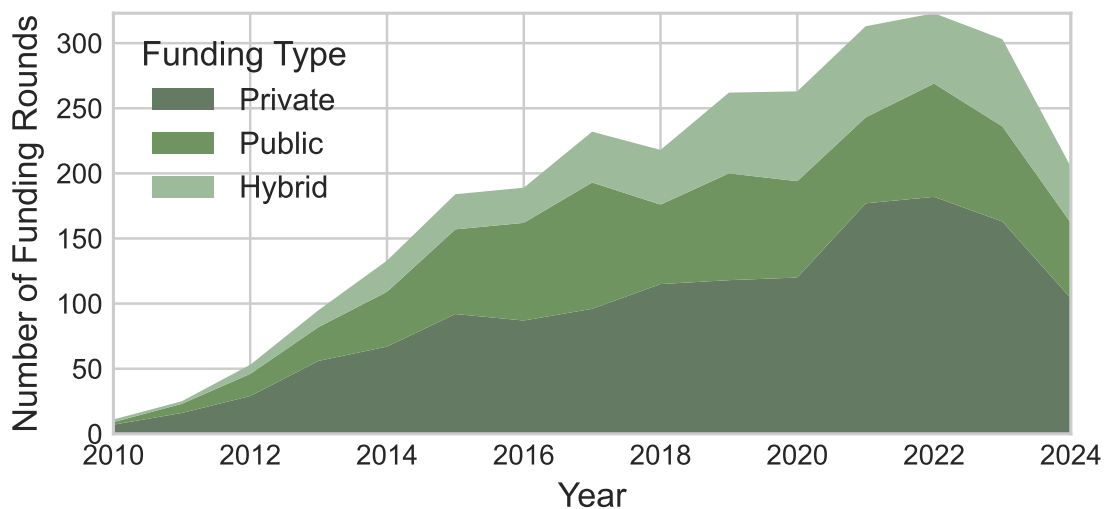


Figure 4.5: Share of Funding Rounds over Time

4.1.3 Investors

This final descriptive section provides a high-level overview of the 2,716 unique investors who participated in the funding rounds. Figure 4.6 illustrates the geographic distribution of these investors, broken down by their classified archetype. The chart reinforces the finding from the startup distribution, showing that investor headquarters are also heavily concentrated in North America and Western Europe. While private investors represent the largest archetype in both of these dominant regions, the data reveals a notable dis-

tion. In Western Europe, public and hybrid investors constitute a proportionally more significant part of the ecosystem, suggesting a more established landscape for blended finance models compared to other regions.

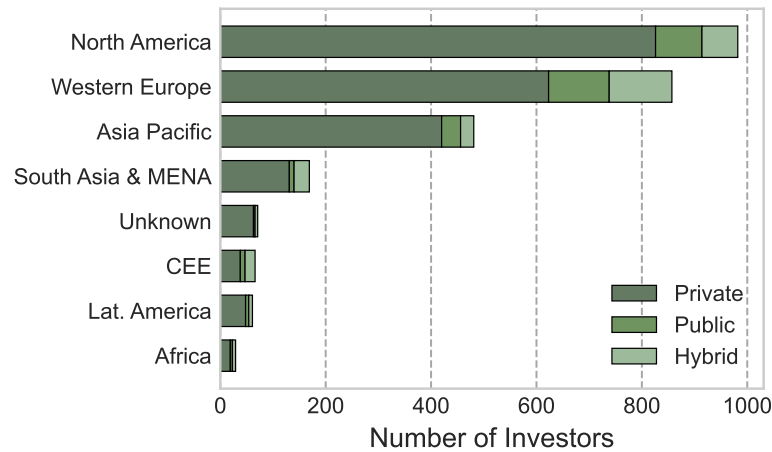


Figure 4.6: Distribution of Investor Headquarter Locations by Type

Lastly, table 4.6 details the activity level of these investors within the sample. The data reveals a highly skewed distribution: while the mean number of investments per investor is 1.87, the median is just 1.00. This indicates that at least half of all investors in the dataset participated in only a single funding round. This "long tail" phenomenon suggests the sustainable investment landscape consists of many infrequent or single-deal participants alongside a small core of highly active, repeat investors, with the most active entity making 153 investments.

Table 4.6: Descriptive Statistics of Investor Activity in the Sample

	Mean	Median	Std. Dev.	Min	Max
Investments per Investor	1.87	1.00	4.42	1	153

4.2 Findings for H1: Follow-on Funding

4.2.1 Descriptive Analysis

This section begins the empirical investigation of Hypothesis 1, which posits that sustainable startups initially receiving hybrid funding are more likely to secure subsequent rounds of follow-on funding. The theoretical reasoning for this hypothesis is rooted in the concept of dual validation: the presence of a public co-investor signals that the venture's technology and societal impact claims have been vetted, while the private co-investor provides crucial market validation of a commercially viable business model. This unique combination is expected to create a comprehensively de-risked and compelling opportunity for later-stage investors.

To test this, the analysis is structured in three parts. It first tests the primary hypothesis on the likelihood of securing any follow-on funding. This is followed by two post-hoc analyses that delve deeper: one exploring the impact on the financial amount of follow-on funding, and a second investigating the moderating role of the investment's nature (i.e., dilutive vs. non-dilutive). Before presenting the results of these models, the following subsection provides a descriptive overview of the key variables.

Of the 1,638 startups included in the analysis 564 (34.4%) successfully raised at least one funding round after their initial investment. The remaining 1,074 startups (65.6%) did not secure further funding within the observation period.

Table 4.7 presents a crosstabulation of initial investor type against the likelihood of receiving follow-on funding. The descriptive data reveals a clear difference in outcomes. Startups that began with a Private round had the highest success rate, with 37.1% securing subsequent financing, followed closely by startups with a Hybrid first round at 36.3%. In stark contrast, startups that received Public funding in their initial round were least likely to secure further investment, with only 28.8% doing so. These initial figures present a nuanced picture in relation to H1. While not showing a descriptive advantage for hybrid-funded startups over privately-funded ones, the data strongly suggests that the presence of a private, market-oriented investor—either alone or in a hybrid syndicate—is a powerful positive signal for subsequent funders. This dynamic will be explored more rigorously in the multivariate analysis to follow

Table 4.7: Follow-on Funding Success Rate by Initial Funding Type

Initial Funding Type	Received Follow-on Funding?		Total
	No	Yes	
Private	508 (62.9%)	299 (37.1%)	807 (100%)
Public	348 (71.2%)	141 (28.8%)	489 (100%)
Hybrid	218 (63.7%)	124 (36.3%)	342 (100%)
Total	1074 (65.6%)	564 (34.4%)	1638 (100%)

Note: Values represent startup counts, with row percentages in parentheses.

To provide context for the subsequent regression analysis, table 4.8 summarizes the key characteristics of the initial funding round across the three investor archetypes. The data shows that the timing of the first investment is broadly consistent, with startups being of a similar age (around 4 years old) and receiving funding in a similar timeframe (mean year 2018) regardless of the investor type.

In stark contrast, the financial scale of these initial rounds differs dramatically. A private first round is substantially larger (median: \$1.75 million) than a hybrid (median: \$0.70 million) or public round (median: \$0.21 million). For all archetypes, the significant gap

between the mean and median funding amounts underscores the right-skewed nature of venture finance, where averages are heavily influenced by a few exceptionally large deals. These factors will be controlled for in the subsequent model to isolate the effect of the initial investor type.

Table 4.8: Descriptive Statistics of First Round Characteristics by Archetype

Statistic	Private	Public	Hybrid
Age at First Round (Years)			
Mean	4.13	4.29	4.27
Median	3.51	3.79	3.68
First Round Funding Amount (\$)			
Mean	37,168,664	9,051,371	12,068,710
Median	1,747,939	208,929	701,796
Std. Dev.	137,415,690	78,192,641	90,360,600
First Round Funding Year			
Mean	2018	2018	2019
Median	2019	2018	2019

4.2.2 Primary Analysis (H1): Likelihood of Securing Follow-on Funding

To test the primary hypothesis on the likelihood of securing follow-on funding, a binary logistic regression was employed. This model is a form of Generalized Linear Model well-suited for binary outcomes; it uses a binomial family with a logit link function to model the probability of an event's occurrence. The results in Table 4.9 are presented as coefficients in log-odds, which are then converted into more intuitive Odds Ratios (OR). An OR greater than 1 indicates increased odds of the outcome, an OR less than 1 indicates decreased odds, and statistical significance is denoted using standard p-value thresholds. This primary analysis is followed by two post-hoc analyses that explore further nuances of the funding relationship.

Table 4.9 presents the results of this analysis in a stepwise manner, comparing two nested models to isolate the effect of the initial investor type. Model 1 serves as a baseline, containing only the control variables: the startup's age at its first round, the size of that round, and fixed effects for industry and geographic region. Model 2 builds on this baseline by introducing the primary independent variable, **First Round Funding Type**, using *Hybrid* funding as the reference category for comparison.

Table 4.9: Logistic Regression Results for Likelihood of Follow-on Funding

Variable	Model 1 (Controls Only)		Model 2 (+ Funding Type)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Independent Variable (Ref: Hybrid)				
Private			-0.0380 (0.96)	(0.145)
Public			-0.3995** (0.67)	(0.160)
Control Variables				
Age at First Round (Years)	-0.1590*** (0.85)	(0.025)	-0.1547*** (0.86)	(0.025)
First Round Amount (\$M)	-0.0013* (0.001)	(0.001)	-0.0016* (0.001)	(0.001)
First Round Year	-0.0008 (0.020)	(0.020)	-0.0048 (0.020)	(0.020)
Region (dummies)	Yes		Yes	
Industry (dummies)	Yes		Yes	
Intercept	1.3621	(40.497)	9.5447	(40.778)
Model Fit				
Observations	1,638		1,638	
Log-Likelihood	-991.46		-986.83	
Pseudo R ²	0.074		0.079	

Note: Coefficients are log-odds with OR in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster dummy variables were included but not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The baseline model (Model 1) reveals that even after accounting for fixed effects, a startup's age at its first funding round is a highly significant predictor. For each additional year before securing initial funding, the odds of receiving follow-on investment decrease by approximately 15% (OR = 0.85, $p < 0.01$), suggesting that investors tend to favor younger ventures.

Building on this, Model 2 introduces the initial funding type to directly test Hypothesis 1. The findings reveal a distinct hierarchy of effectiveness. There is no statistically significant difference in the odds of securing follow-on funding between startups with an initial Private round and those in the Hybrid reference group. However, the coefficient for Public funding is negative and highly significant ($p < 0.05$). The odds ratio of 0.67 indicates that, holding all other factors constant, a startup that begins with a purely Public round is approximately 33% less likely to receive follow-on funding compared to one that begins with a Hybrid round.

The LRT, assessing whether the addition of the investor-type variables provides a statistically significant improvement to the model's fit, confirms that the more complex model

(Model 2) is superior ($\chi^2(2) = 9.26, p < 0.01$).

In summary, these results offer only partial support for Hypothesis 1. While a hybrid start is an effective pathway to future investment, the model shows it holds no statistical advantage over a purely private start. The findings instead suggest that the more decisive factor is the presence of a private, market-oriented investor. Both pathways that include this component — purely private and hybrid — are statistically equivalent in their effectiveness and are significantly more likely to lead to follow-on funding than a solely public first round.

4.2.3 Post-Hoc Analysis I (H1): Amount of Follow-on Funding

The primary analysis for H1 demonstrated that startups with an initial Private or Hybrid round were significantly more likely to secure any follow-on funding compared to those with a purely Public start. This raises a subsequent question: does the influence of the initial funding type extend beyond the mere likelihood of future funding to the actual amount of capital raised?

To answer this question, Table 4.10 presents the relevant descriptive statistics. A meaningful analysis of financial scale requires focusing exclusively on the 564 startups that successfully secured subsequent capital, as the full sample is skewed by the large number of firms with no follow-on funding rounds and hence 0 follow-on funding. Among this successful subset, a clear hierarchy emerges. Startups initially receiving private funding not only raised the highest mean amount (\$38.6 million) but also the largest median amount (\$3.1 million). Interestingly, while startups with a public start raised a higher mean amount than hybrid-started ventures (\$13.6 million vs. \$7.6 million), their median funding was the lowest of the three groups at just \$0.75 million. This suggests that while a few publicly-backed startups secure large follow-on rounds, the typical hybrid-backed venture raises more capital in its subsequent rounds.

Table 4.10: Descriptive Statistics of Follow-on Funding Amount by Initial Funding Type

Statistic	Private	Public	Hybrid
N (with Follow-on)	299	141	124
Mean Follow-on Amount	\$38,608,139	\$13,642,921	\$7,641,216
Median Follow-on Amount	\$3,069,053	\$750,000	\$1,207,208

Note: Statistics are calculated only for the subset of 564 startups that received at least one round of follow-on funding. All monetary amounts are in USD.

To formally test these observed differences while controlling for other factors, an OLS regression was performed. The model uses the log-transformed **follow-on amount** as the continuous dependent variable to account for the significant right-skew in the funding data. Consequently, the coefficients presented in Table 4.11 represent the change in the log of the follow-on amount for a one-unit change in a predictor. Following the primary

analysis, the results are presented using a nested model approach in the subsequent table.

Table 4.11: OLS Regression Results for Log-Transformed Follow-on Amount

Variable	Model 1 (Controls Only)		Model 2 (+ Funding Type)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Independent Variable (Ref: Hybrid)				
Private Investor			0.2326***	(0.073)
Public Investor			−0.1569**	(0.079)
Control Variables				
First Round Amount (\$M)	0.0007***	(0.000)	0.0006**	(0.000)
First Round Year	0.0175*	(0.010)	0.0155	(0.010)
Age at First Round	−6.1366*	(3.686)	−5.0610	(3.654)
Industry Dummies	Yes		Yes	
Region Dummies	Yes		Yes	
Intercept	−35.1393	(20.016)	−31.1729	(19.882)
Model Fit				
Observations	1638		1638	
R ²	0.033		0.056	
Adj. R ²	0.024		0.046	
F-test vs. Prev. Model (p-value)	-		< 0.001	

Note: Standard errors are in parentheses. The dependent variable is the log-transformed follow-on funding amount. Regional and industry cluster dummy variables were included but are not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The results of the nested OLS regression, presented in Table 4.11, reveal a clear hierarchy regarding the financial magnitude of follow-on funding. The full model (Model 2) - which offers a statistically significant improvement in explanatory power over the baseline ($F(2, 1620) = 19.46, p < .001$) — shows that a purely Private start is the most financially advantageous path. Startups on this track raise significantly more follow-on capital than their hybrid-backed peers ($p < 0.01$). This finding contrasts with the earlier analysis of likelihood, where both pathways performed equally. Meanwhile, consistent with previous results, an initial Public round remains the least advantageous path, associated with raising significantly less follow-on capital than the Hybrid reference group.

Interestingly, the influence of the control variables shifts compared to the primary analysis. While the size of the initial funding round remains a significant predictor, the startup's age at its first round loses its statistical significance in this model. This suggests that while a startup's age may affect its likelihood of securing any follow-on funding, it does not appear to predict the financial scale of that funding for the ventures that succeed.

This overall finding — that a purely private start leads to the greatest financial outcome

— runs contrary to the spirit of the original hypothesis. It suggests that while the “dual validation” of a hybrid round is equally effective as a private round for opening the door to subsequent funding, the singular, market-driven focus of a purely private backer is the strongest catalyst for maximizing the financial magnitude of that future funding.

4.2.4 Post-Hoc Analysis II (H1): The Moderating Role of Funding Nature

The preceding analyses have consistently shown that an initial public funding round is associated with a lower likelihood and amount of follow-on funding. This finding is particularly striking given that public funding is often explicitly designed to bridge early-stage funding gaps and catalyze private investment. This apparent contradiction raises a crucial question about the nature of the public funding signal: is the “Public” category monolithic, or does its effect depend on the structure of the investment—specifically, whether capital is provided as dilutive equity or as a non-dilutive grant?

To investigate this, a final post-hoc analysis was conducted, introducing an interaction term between the initial investor type and the funding’s nature. Before examining the model results, it is useful to understand the distribution of these investment structures across the archetypes, as shown in Table 4.12. The data reveals starkly different preferences: both Private and Hybrid rounds are predominantly dilutive, reflecting a market-oriented focus on equity. In sharp contrast, Public funding is overwhelmingly non-dilutive, consistent with a state logic of providing support without taking an ownership stake. This fundamental difference is central to understanding the interaction tested in the following model.

Table 4.12: Distribution of First Round Funding Type and Funding Nature

Funding Nature	dilutive	non-dilutive
Hybrid	262	80
Private	681	126
Public	77	412

To provide an intuitive overview of these combined effects before examining the regression coefficients, Figure 4.7 visualizes the predicted probabilities of securing follow-on funding. These probabilities are generated from the full interaction model for each of the six distinct pathways (e.g., Private-Dilutive, Public-Non-dilutive), holding all other control variables at their mean.

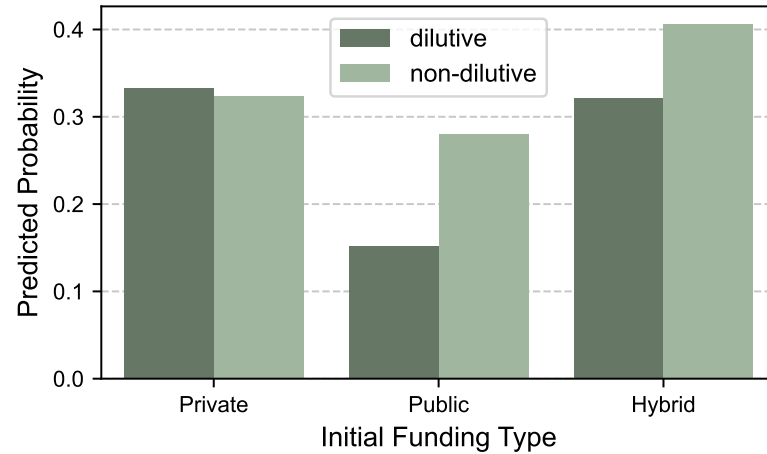


Figure 4.7: Predicted Probability of Follow-on Funding by Initial Funding

The plot immediately suggests that the nature of the investment is a crucial moderator of success. For both the Private and Hybrid archetypes, the performance difference between their dilutive and non-dilutive paths appears less pronounced, although the Hybrid-Non-dilutive path yields the highest probability of success overall. Most strikingly, the visualization provides a clear answer to this section's opening question: public funding is not monolithic. There is a dramatic performance gap between a non-dilutive public round, which is highly effective, and a dilutive one, which is the least effective pathway of all six. This suggests the structure of public capital, not just its source, is a critical determinant of its signaling power to future investors.

To formally test these visual patterns, Table 4.13 compares the main effects model (Model 2) from the primary analysis with the full interaction model (Model 3). This analysis again uses a binary logistic regression, with results presented in log-odds and their corresponding Odds Ratios. For this interaction model, the *Hybrid* type combined with a *Dilutive* nature serves as the reference category against which all other combinations are compared.

Table 4.13: Comparing Main Effects and Interaction Models for Follow-on Funding

Variable	Model 2 (Main Effects)		Model 3 (Interaction Model)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Independent Variable (Ref: Hybrid)				
Private	−0.0380 (0.96)	(0.145)	0.0502 (1.05)	(0.163)
Public	−0.3995** (0.67)	(0.160)	−0.9751*** (0.38)	(0.330)
Moderator (Ref: Dilutive)				
Non-dilutive			0.3662 (1.44)	(0.280)
Interaction Terms				
Private : Non-dilutive			−0.4077 (0.358)	
Public : Non-dilutive			0.4095 (0.424)	
Control Variables				
Age at First Round (Years)	−0.1547*** (0.025)		−0.1584*** (0.026)	
First Round Amount (\$M)	−0.0016* (0.001)		−0.0016* (0.001)	
First Round Year	−0.0048 (0.020)		−0.0063 (0.020)	
Region (dummies)	Yes		Yes	
Industry (dummies)	Yes		Yes	
Intercept	9.5447 (40.778)		12.3265 (40.991)	
Model Fit				
Observations	1,638		1,638	
Log-Likelihood	−986.83		−982.77	
Pseudo R ²	0.079		0.084	

Note: Coefficients are log-odds with OR in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster dummy variables were included but not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The full interaction model (Model 3) reveals a more complex dynamic, where the effect of the investor type depends on the investment's nature. With the 'Hybrid-Dilutive' pathway serving as the reference category, the main effect coefficients now compare the other dilutive pathways to this baseline. The significant negative coefficient for 'Public' (−0.9751, $p < 0.01$), for example, indicates that an initial dilutive public round is far less likely to attract follow-on funding than a dilutive hybrid round. While the individual interaction terms lack significance on their own, their collective impact provides the key insight.

The importance of this more complex model is confirmed by a LRT, which shows a statistically significant improvement in model fit over the main-effects-only model (Model 2) ($\chi^2(3) = 8.12, p = 0.0436$). This result validates the presence of a moderating effect and confirms that the relationship between investor type and follow-on funding is not uniform

across all conditions. Because interpreting the full effect for each of the six pathways directly from these coefficients is not intuitive, a full set of post-hoc pairwise comparisons is presented next, to clearly delineate the differences between each distinct pathway.

To precisely identify which of the six funding pathways differ from one another, a full set of post-hoc pairwise comparisons was conducted using Wald tests. The results are presented in Table 4.14. Each cell in the table shows a direct comparison between the 'Reference' group (the row) and the 'Comparison Group' (the column). The top value is the difference in log-odds, which is converted into a more intuitive OR in parentheses below. Again, an OR greater than 1 means the row group has higher odds of success (= raising follow-on funding) than the column group.

Table 4.14: Pairwise Comparisons of Predicted Follow-on Funding Likelihood

Reference	Comparison Group					
	Hybrid (D)	Hybrid (ND)	Private (D)	Private (ND)	Public (D)	Public (ND)
Hybrid (D)	-	-0.37 (0.69)	-0.05 (0.95)	-0.01 (0.99)	0.98*** (2.66)	0.20 (1.22)
Hybrid (ND)	0.37 (1.45)	-	0.32 (1.38)	0.36 (1.43)	1.34*** (3.82)	0.57 (1.77)
Private (D)	0.05 (1.05)	-0.32 (0.73)	-	0.04 (1.04)	1.03*** (2.80)	0.25 (1.28)
Private (ND)	0.01 (1.01)	-0.36 (0.70)	-0.04 (0.96)	-	0.99*** (2.69)	0.21 (1.23)
Public (D)	-0.98*** (0.38)	-1.34*** (0.26)	-1.03*** (0.36)	-0.99*** (0.37)	-	-0.78** (0.46)
Public (ND)	-0.20 (0.82)	-0.57 (0.57)	-0.25 (0.78)	-0.21 (0.81)	0.78** (2.18)	-

Note: Cells contain the difference in log-odds between the row group and the column group (Row - Column), with Odds-Ratios in parentheses. (D) = Dilutive, (ND) = Non-dilutive. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

This comprehensive comparison clarifies the findings significantly, revealing a three-tiered hierarchy of effectiveness:

1. The Top Tier: The first key finding is that there are no statistically significant differences among any of the four pathways involving private capital: Private-Dilutive, Private-Non-dilutive, Hybrid-Dilutive, and Hybrid-Non-dilutive. These four approaches are statistically indistinguishable from one another in their strong ability to attract subsequent funding.
2. The Negative Outlier: In stark contrast, the Public-Dilutive pathway is a significant outlier. It performs statistically worse than all four top-tier pathways, with its odds of

success being just 26% to 38% of theirs. This confirms that an initial public equity investment acts as a negative signal to later-stage investors.

3. The Powerful Public Signal: Most importantly, the table reveals the nuanced position of the Public-Non-dilutive pathway. While it is significantly more effective than its dilutive public counterpart ($OR = 2.18, p < 0.05$), it is statistically indistinguishable from the four top-tier private and hybrid pathways. This suggests that a non-dilutive public grant can act as a powerful and "clean" signal of quality, effectively launching a startup onto a funding trajectory equivalent to one backed by market-oriented investors.

4.3 Findings for H2: Exit

Having explored the factors influencing a startup's ability to secure follow-on funding, the analysis now shifts to a second, more definitive measure of success: a commercial exit. This section begins the empirical investigation of Hypothesis 2, which posits that a greater prevalence of private funding in a startup's financing history increases both the likelihood of achieving an exit and the speed at which it occurs. The theoretical reasoning for this hypothesis is rooted in the market logic of private investors. Their primary goal is to realize financial returns through a liquidity event, and their non-monetary support—from strategic guidance to network access—is therefore deliberately targeted to accelerate a startup's path toward a profitable acquisition or IPO.

4.3.1 Descriptive Analysis

The analysis for this hypothesis includes the full sample of 1,638 sustainable startups. Of these, 117 (7.1%) achieved a successful exit during the observation period. The remaining 1,521 startups (92.9%) had not exited by the end of 2024.

To understand the preliminary relationship between funding composition and exit likelihood, table 4.15 provides a comparative overview of the funding histories for startups that exited versus those that did not.

Table 4.15: Descriptive Statistics of Startups by Exit Status

Exit Status	No (N = 1521)	Yes (N = 117)
Funding Round Composition		
Ratio Private Rounds	0.48	0.68
Ratio Public Rounds	0.32	0.13
Ratio Hybrid Rounds	0.20	0.18
Funding Activity & Scale		
No. Funding Rounds	1.70	1.81
Total Funding (\$M)	44.05	76.82
Age at Exit (Years)	-	7.67

Note: All metrics are the calculated Mean across the category

The descriptive data in Table 4.15 reveals a clear divergence between the two groups.

- *Funding Composition:* The funding portfolios of the two groups differ significantly. For startups that exited, private funding rounds constitute a clear majority, making up 68% of their funding history on average. In contrast, the portfolios of non-exited startups are more balanced, with a notably lower proportion of private rounds (48%) and a significantly higher share of public rounds (32% vs. 13% for exited firms).
- *Financial Scale:* A significant disparity in financial scale is also evident. Exited startups raised, on average, nearly 75% more total funding (\$76.82M) than those that did not exit (\$44.05M).

To investigate the second part of H2 - the speed of exit - fig. 4.8 visualizes the distribution of time-to-exit for the 117 exited startups. The startups are grouped by their dominant funding type, which was determined by the highest ratio funding rounds per type; in case of a tie, the type of the last funding round was chosen. Of the exited startups, 81 were private-dominated, 18 were public-dominated, and 18 were hybrid-dominated.

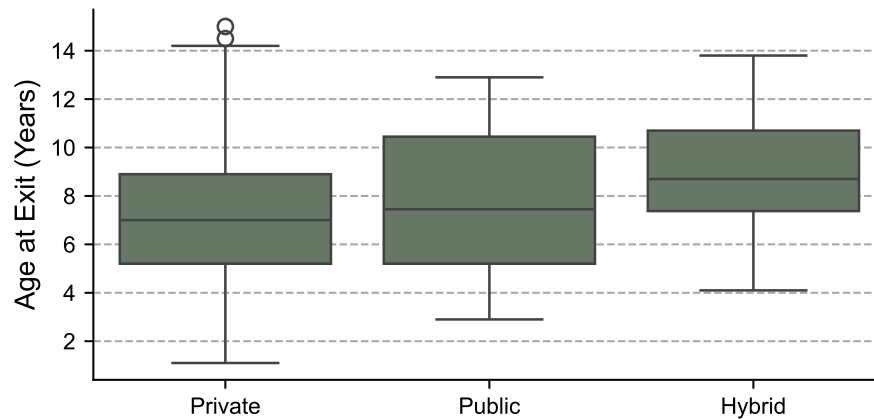


Figure 4.8: Distribution of Startup Age at Exit by Dominant Funding Type

The analysis of the exit timelines reveals a clear relationship between a startup's funding composition and its speed of exit. The box plot highlights two key conclusions:

1. *Private-Dominated Startups Exit Fastest:* The median age at exit is lowest for startups where private funding was the dominant source, at approximately 7 years. This is visibly quicker than for startups dominated by public rounds (median of 7.5 years) and notably faster than those dominated by hybrid rounds, which have the highest median exit age at roughly 8.5 years. This provides strong descriptive support for the hypothesis that a prevalence of private funding is associated with a faster exit.
2. *The Private Path Shows Higher Variability:* While being the fastest on average, the private-dominated path also exhibits the greatest variability. The whisker for this group extends to the youngest exit age (just over 1 year), but also includes an outlier at the oldest age (around 15 years). This suggests that while the private funding route can lead to very rapid exits, it may also contain a wider range of outcomes compared to the hybrid path, which appears more concentrated and predictable, albeit slower.

These descriptive findings provide strong initial support for both parts of Hypothesis 2. Startups with a greater prevalence of private funding appear more likely to exit and, on average, do so more quickly. The following section will use multivariate logistic and Cox regression models to test these relationships formally while controlling for other confounding factors.

4.3.2 Primary Analysis (H2): Likelihood of Exit

To test the first part of Hypothesis 2 on the likelihood of an exit, once more binary logistic regression analysis was conducted. To best operationalize the "prevalence" of each funding type, three different model specifications were tested using (a) the count of rounds, (b) the financial magnitude, and (c) the proportional ratio of rounds for each archetype. The final model was selected based on the AIC, which estimated the relative quality of

the statistical models by balancing its goodness of fit with its complexity. During this assessment, the model using the proportional ratios of funding rounds yielded the lowest AIC score, indicating the best predictive fit, and is therefore presented in Table 4.16. The conducted LRT confirms that this model provides a statistically significant improvement over a baseline containing only controls ($\chi^2(2) = 23.20, p < 0.001$)

Table 4.16: Logistic Regression Results for Likelihood of Startup Exit

Variable	Model 1 (Controls Only)		Model 2 (+ Funding Ratios)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Independent Variables (Ref: Private)				
Ratio Public Rounds			-1.360*** (0.257)	(0.315)
Ratio Hybrid Rounds			-0.207 (0.813)	(0.288)
Control Variables				
Age at Last Round (Years)	-0.166*** (0.847)	(0.036)	-0.164*** (0.849)	(0.035)
No. of Funding Rounds	0.148** (1.160)	(0.071)	0.158** (1.171)	(0.074)
Total Funding (\$M)	0.000 (1.00)	(0.000)	0.000 (1.000)	(0.000)
Last Round Amount (\$M)	0.001 (1.001)	(0.001)	0.001 (1.001)	(0.001)
Industry Dummies	Yes		Yes	
Region Dummies	Yes		Yes	
Founding Year Dummies	Yes		Yes	
Intercept	-2.609** (1.098)		-2.1527* (1.109)	
Model Fit				
Observations	1,638		1,638	
Log-Likelihood	-383.89		-372.29	
Pseudo R ²	0.045		0.058	

Note: Coefficients are log-odds with OR in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster and founding year dummy variables were included but not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The analysis begins with the baseline model (Model 1), which includes only the control variables. The results show that age at last round is a highly significant negative predictor ($p < 0.001$), indicating that a startup's odds of exiting diminish the longer it has been since its last funding event. This suggests that momentum and continuous investor validation are critical. Conversely, the total number of funding rounds is a significant positive predictor ($p < 0.05$), aligning with the expectation that each successful round moves a startup

closer to the scale required for an exit.

Model 2 then introduces the funding ratio variables to test the central hypothesis, using the proportion of private rounds as the baseline for comparison:

- *A High Proportion of Public Funding Significantly Lowers Exit Likelihood.* The model's strongest and most significant finding is the negative coefficient for Ratio of Public Rounds (-1.360 , $p < 0.001$). The corresponding odds ratio of 0.257 ($e^{-1.360}$) can be interpreted as follows: a startup funded entirely by public rounds has only 26% of the odds of achieving a commercial exit compared to a startup funded entirely by private rounds, holding all other factors constant.
- *No Significant Difference Between Hybrid and Private Funding Proportions.* The coefficient for Ratio of Hybrid Rounds (-0.207) is not statistically significant. This indicates that there is no statistical difference in the likelihood of an exit between a startup with a high proportion of hybrid funding and one with a high proportion of private funding. This suggests that the market-oriented discipline present in hybrid funding aligns these ventures on a similar exit trajectory as their purely private-backed peers.

In perspective, these findings strongly support the core hypothesis in a nuanced way. The analysis demonstrates that the relative balance of a startup's funding portfolio is a powerful predictor of its commercial fate. The most critical factor is the significant negative association between a heavy reliance on public funding and the likelihood of an exit. This is consistent with the theoretical framework that public investors, guided by a state logic emphasizing long-term innovation over immediate financial returns, foster a path that is fundamentally different from the exit-focused trajectory driven by the market logic inherent in both private and hybrid funding structures.

4.3.3 Secondary Analysis (H2): Speed of Exit

While the preceding analysis established if a startup is likely to exit, this section addresses the second part of Hypothesis 2: when that exit occurs. The analysis now shifts from likelihood to the speed of exit, investigating the time-to-event.

To model this, a CoxPH model was employed. This survival model is a strong fit for this context as it can handle "censored" data — in this case, the many startups that had not yet exited by the end of the observation period. The model analyzes the duration from a startup's founding to an exit event, using a binary indicator to distinguish between startups that exited and those that were censored. The results are presented as Hazard Ratios (HR). An HR greater than 1 indicates an increased "hazard" or rate of exiting (a faster exit), while an HR less than 1 indicates a reduced hazard (a slower exit).

The results of the nested CoxPH models are presented in Table 4.17. A likelihood-ratio test confirms that adding the funding ratio variables in Model 2 provides a statistically significant improvement over the baseline model of controls ($\chi^2(2) = 23.20$, $p < 0.0001$).

Table 4.17: CoxPH Model Results for Speed of Startup Exit

Variable	Model 1 (Controls Only)		Model 2 (+ Funding Ratios)	
	Coef. (HR)	Std. Err.	Coef. (HR)	Std. Err.
Independent Variables (Ref: Private)				
Ratio Public Rounds			−1.360*** (0.257)	(0.315)
Ratio Hybrid Rounds			−0.207 (0.813)	(0.288)
Control Variables				
Startup Founded Year	0.620* (1.868)	(0.315)	0.630* (1.878)	(0.318)
Age at Last Round (Yrs)	0.460 (1.584)	(0.314)	−0.164*** (0.849)	(0.035)
No. of Funding Rounds	0.140** (1.150)	(0.065)	0.158** (1.171)	(0.074)
Last Round Year	−0.620* (0.538)	(0.315)	−0.630* (0.533)	(0.318)
Total Funding (\$M)	0.000 (1.000)	(0.000)	0.000 (1.000)	(0.000)
Last Round Amount (\$M)	0.000 (1.000)	(0.000)	0.001 (1.001)	(0.001)
Industry Dummies	Yes		Yes	
Region Dummies	Yes		Yes	
Model Fit				
Partial Log-Likelihood	-789.27		-372.29	
Concordance	0.71		—	
Partial AIC	1614.53		800.58	

Note: Coefficients are log(Hazard Ratio) with HR in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster dummy variables were included but not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Examining the baseline model (Model 1), the No. of Funding Rounds is a significant predictor of a faster exit (HR = 1.15). Interestingly, the Age at Last Round, which was a strong predictor of the likelihood of an exit, is not a significant predictor of the speed of exit, suggesting that while a long time since the last round makes an exit less likely, it doesn't slow the timeline for those that do exit.

Introducing the funding composition variables in Model 2 reveals their powerful influence on the exit timeline. The most significant finding is the strong negative coefficient for the Ratio of Public Rounds. The Hazard Ratio of 0.257 suggests that a startup funded entirely by public rounds has a 74.3% lower rate of exiting at any point in time compared

to a purely privately funded venture. This strongly supports the hypothesis that public funding is associated with a significantly longer, more patient trajectory. In contrast, the coefficient for the Ratio of Hybrid Rounds is not statistically significant, indicating no clear difference in exit speed compared to the private-round baseline.

The Hazard Ratios from the full model are visualized in the forest plot in Figure 4.9. This plot provides a clear summary of each variable's effect, where each point is the Hazard Ratio and the horizontal line is its 95% confidence interval. If a confidence interval crosses the vertical line at $HR = 1$, the effect is not statistically significant. The plot starkly illustrates the main finding: the point and confidence interval for Ratio of Public Rounds are far to the left of 1, indicating a statistically significant reduction in the hazard of an exit. Conversely, the intervals for most other variables, including Ratio of Hybrid Rounds, clearly overlap with 1.

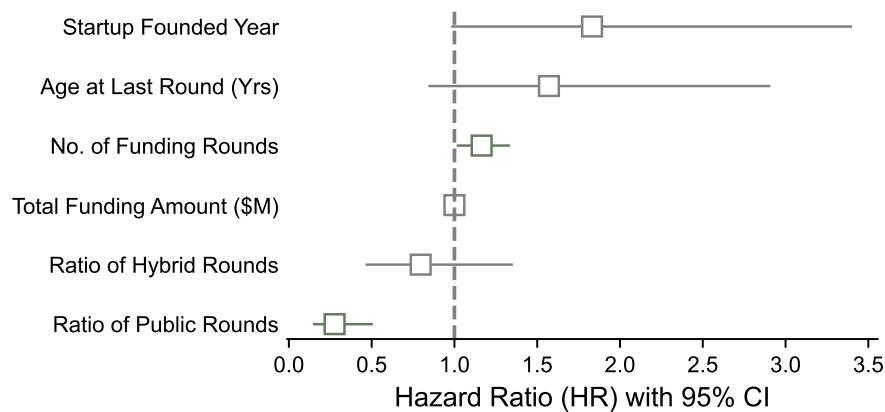


Figure 4.9: Forest Plot of Hazard Ratios for Speed of Exit

Note: Figure displays Hazard Ratios and 95% confidence intervals. An $HR > 1$ indicates a faster exit, while an $HR < 1$ indicates a slower exit.

Overall, the analysis supports H2, highlighting a statistically significant link between a high prevalence of public capital and longer exit horizons, while providing some suggestive — but inconclusive — evidence regarding hybrid funding.

4.4 Findings for H3: Long-Term Survival

Having examined success through the market-oriented lenses of follow-on funding and commercial exits, the final analysis turns to a third, distinct success metric: long-term survival. This section investigates Hypothesis 3, which explores whether a startup's funding composition influences its ability to endure over time. The hypothesis posits that the "patient capital" associated with public funding, driven by a state logic that prioritizes long-term impact over rapid returns, provides a crucial layer of stability. By insulating ventures from purely market-driven pressures, this type of funding is expected to enhance a startup's ability to remain operational and pursue its mission over a longer time horizon.

4.4.1 Descriptive Analysis

To test for long-term operationality, the analysis focuses on the filtered subsample of 905 mature startups that had reached a minimum age of 10 years by the end of the observation period. Of these firms, 687 (75.9%) were verified as still operating, while the remaining 218 (24.1%) were confirmed to have closed. This approach, while necessary to assess long-term outcomes, does introduce a survivor bias, as it only includes firms that have already survived for a decade. This has already been discussed in section 3.6.

A high-level view of this mature cohort is presented in Figure 4.10. The bar chart visualizes the survival rates for startups grouped by their dominant funding type. While the differences are not dramatic, the descriptive data shows a slight trend in favor of the hypothesis: public- (76%) and hybrid-dominated (78%) firms exhibit a marginally higher rate of long-term operationality compared to their private-funded peers (75%).

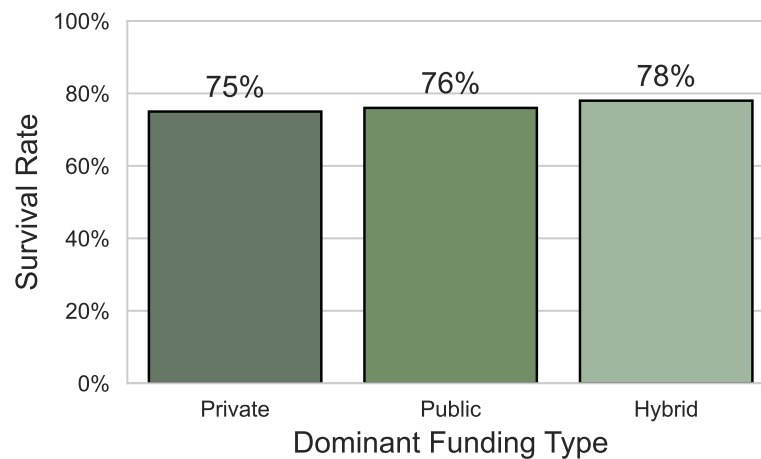


Figure 4.10: Long-Term Survival Rate by Dominant Funding Type

Only startups aged 10+ years were included

A more detailed breakdown, presented in Table 4.18, reinforces this subtle pattern. The funding composition of operating startups shows a slightly higher proportion of public rounds (34%) compared to firms that closed (31%). The differences in funding mix are modest, but the disparity in financial scale is more pronounced, with operating firms having raised substantially more capital over their lifetime.

Table 4.18: Descriptive Statistics of Startups by Operating Status (in 2025)

Survival Status	Closed (N = 218)	Operating (N = 687)
Funding Round Composition (Ratio)		
Ratio Private Rounds	0.52	0.48
Ratio Public Rounds	0.31	0.34
Ratio Hybrid Rounds	0.16	0.18
Funding Activity & Scale		
Total Funding (\$M)	15.63	51.04
No. of Funding Rounds	1.53	1.75
Age at Last Round ((Years)	3.45	6.76

Note: All metrics are the calculated Mean across the category

While these descriptive statistics provide some preliminary, albeit slight, directional support for the hypothesis, it is unclear if these patterns are statistically significant. The following section therefore employs a logistic regression analysis to formally test whether the prevalence of public funding has a significant impact on long-term survival after controlling for other factors.

4.4.2 Primary Analysis (H3): Long-Term Survival

Consistent with the methods used for the preceding hypotheses, a nested binary logistic regression was employed to formally test the link between funding composition and long-term survival. An initial exploration revealed that while funding ratios were effective for predicting exits, the financial amount from each archetype was a more significant predictor for survival. Consequently, the analysis was structured around this variable and proceeds in three nested steps, presented across Table 4.19 and Table 4.20

- **Model 1** establishes a baseline using only control variables, including the total funding amount.
- **Model 2** disaggregates total funding into the amounts received from private, public, and hybrid sources to see if the source of capital matters.
- **Model 3** further disaggregates these amounts by their nature (dilutive vs. non-dilutive) to test for interaction effects.

Table 4.19: Logistic Regression Results for Long-Term Survival (by Funding Amount)

Variable	Model 1 (Controls Only)		Model 2 (+ Archetype)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Funding Variables				
Total Amount (\$M)	0.0031** (1.003)	(0.001)		
Private Amount (\$M)			0.0025** (1.003)	(0.001)
Public Amount (\$M)			0.0296 (1.030)	(0.020)
Hybrid Amount (\$M)			0.1506** (1.163)	(0.068)
Control Variables				
Number of Funding Rounds	0.0820 (1.085)	(0.067)	0.0325 (1.033)	(0.068)
Founding Year	-0.0601 (0.942)	(0.048)	-0.0557 (0.946)	(0.049)
Industry Dummies	Yes		Yes	
Region Dummies	Yes		Yes	
Intercept	122.2261	(97.351)	113.1173	(98.135)
Model Fit				
Observations	905		905	
Log-Likelihood	-483.28		-475.61	
Pseudo R ²	0.036		0.052	

Note: Coefficients are log-odds with OR in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster dummy variables were included but are not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The analysis begins with the baseline model in Table 4.19. The results indicate that the Total Amount of funding raised is a significant positive predictor of long-term survival ($p < 0.05$), confirming the intuitive idea that better-capitalized firms are more likely to endure over a decade.

The key question, however, is whether the source of that capital matters more. Model 2 in Table 4.19 addresses this by disaggregating the total funding amount. A LRT confirms this model provides a significantly better fit ($\chi^2(2) = 15.34$, $p < 0.001$), revealing that the source is indeed crucial. While the amount of purely public funding has no significant effect on survival, the amount of both private funding ($p < 0.05$) and hybrid funding ($p < 0.05$) are positively and significantly associated with long-term operationality.

Table 4.20: Logistic Regression Results for Long-Term Survival (by Funding Nature)

Variable	Model 2 (+ Archetype)		Model 3 (+ Archetype & Nature)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Funding Variables				
Private Amount (\$M)	0.0025** (1.003)	(0.001)		
Public Amount (\$M)	0.0296 (1.030)	(0.020)		
Hybrid Amount (\$M)	0.1506** (1.163)	(0.068)		
Private Dilutive Amount (\$M)			0.0023* (1.002)	(0.001)
Public Dilutive Amount (\$M)			0.0054 (1.005)	(0.013)
Hybrid Dilutive Amount (\$M)			0.1319** (1.141)	(0.067)
Private Non-Dilutive Amount (\$M)			0.0050 (1.005)	(0.005)
Public Non-Dilutive Amount (\$M)			0.0924* (1.097)	(0.053)
Hybrid Non-Dilutive Amount (\$M)			5.4593 (234.9)	(4.149)
Control Variables				
Number of Funding Rounds	0.0325 (1.033)	(0.068)	0.0123 (1.012)	(0.069)
Founding Year	-0.0557 (0.946)	(0.049)	-0.0578 (0.944)	(0.049)
Industry Dummies	Yes		Yes	
Region Dummies	Yes		Yes	
Intercept	113.1173	(98.135)	117.1498	(98.776)
Model Fit				
Observations	905		905	
Log-Likelihood	-475.61		-470.54	
Pseudo R ²	0.052		0.062	

Note: Coefficients are log-odds with OR in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster dummy variables were included but are not shown for brevity. Significance levels: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

A three way nested comparison can be found in Appendix A.5

To further pinpoint the drivers of this effect, the final model (Model 3) in Table 4.20 separates the funding amounts by both archetype and nature. The Likelihood Ratio Test is again significant ($\chi^2(3) = 10.15, p < 0.05$), indicating this final level of detail adds crucial

explanatory power. Model 3's results suggest the positive survival effect is driven by two specific pathways: hybrid-dilutive funding ($p < 0.05$) and, most notably, public-non-dilutive funding ($p < 0.1$).

In summary, this deeper analysis provides a nuanced answer to Hypothesis 3. While the simple prevalence of public funding does not predict survival, the financial composition of a startup's history does. The results suggest that the "patient capital" effect of public funding is most pronounced when it is non-dilutive (e.g., grants), providing a stable capital base without complicating governance. At the same time, hybrid funding that includes the market discipline of an equity stake also strongly correlates with endurance, highlighting a separate robust pathway to long-term success.

This chapter has presented the empirical findings from the statistical analyses of the three core hypotheses. The results revealed distinct pathways associated with each funding archetype. The analyses for H1 and H2 demonstrated that private and hybrid funding are most effective for achieving market-oriented milestones, such as securing subsequent investment and accelerating the path to an exit. The analysis for H3, however, painted a more complex picture of long-term survival, where the specific structure of capital appears to be more critical than the funding source alone.

While this chapter has focused on presenting these results objectively, their full meaning remains to be explored. The following chapter will now provide a comprehensive discussion, interpreting these empirical findings through the theoretical framework established earlier and exploring their practical implications for entrepreneurs, investors, and policy-makers.

5 Discussion

The empirical findings have established that different funding archetypes lead sustainable startups along distinct pathways to success. This chapter now addresses the question of why these divergent outcomes occur. To do so, it will interpret the results for each of the three hypotheses through the theoretical lenses of Institutional Logics and Resource Dependence Theory. The discussion will then culminate in a holistic synthesis, weaving the individual findings into a broader narrative about the strategic trade-offs inherent in the funding pathways available to sustainable startups.

5.1 Discussion of Findings for H1: The Power of Investor Signals

The investigation into Hypothesis 1 — which posited that an initial hybrid round would be the most effective pathway to subsequent financing— yielded a more complex reality than anticipated. The findings only partially support this hypothesis, revealing instead that the presence of a market-oriented investor is the most critical signal for attracting future capital. This discussion will first interpret this primary finding, before dissecting the nuanced and contradictory role of public funding that was uncovered in the post-hoc analyses.

The Prevailing Power of Market Logic

The regression analysis showed no statistically significant difference between a private and a hybrid start in securing follow-on funding. Both pathways proved significantly more effective than a solely public round. This suggests that for later-stage investors, the key signal is the presence of a private counterpart who has already performed commercial due diligence and validated the venture's business model. While the 'dual validation' of hybrid funding is theoretically powerful, these results indicate it is the market logic component that prevails in the eyes of future funders. The public partner's role appears to be getting the startup 'ready for market,' but it is the private partner's stamp of approval that makes it truly 'market-ready'.

This interpretation is further bolstered by the first post-hoc analysis, which examined the amount of follow-on funding raised. In that model, a purely private start was not just equal but statistically superior to a hybrid start, leading to significantly larger subsequent funding rounds. This reinforces the conclusion that a strong, undiluted signal of market logic is the most powerful catalyst for attracting future private capital.

The Two Faces of Public Capital

The initial finding that public funding underperforms seems to contradict its core policy objective of bridging funding gaps and catalyzing private investment. This apparent puzzle necessitated the deeper dive in the second post-hoc analysis, which revealed that public funding is not monolithic; its effect depends entirely on its structure.

A non-dilutive public investments acts as a pure and powerful certification signal. In line with Resource Dependence Theory, it provides critical early-stage capital without complicating the ownership structure or governance. This 'clean' signal seemingly makes the venture attractive, elevating its chances of securing future funding to a level statistically indistinguishable from its private- or hybrid-funded peers.

Conversely, a dilutive public investment appears to send a potent negative signal. The market may interpret this as a sign that the startup was unable to attract private capital on its own terms, suggesting a lack of commercial validation. Furthermore, this finding aligns with prior research, which suggests that state-backed ventures can lack market-efficiency. The poor performance may therefore not only stem from a negative market signal, but also from the internal bureaucratic inefficiencies and political pressures often associated with public funding initiatives (Brander, Egan, et al. 2008; Karsai 2018). This pathway proved to be the least effective of all, performing significantly worse than every other funding combination, including its non-dilutive public counterpart. It is important to note, however, that this subgroup was the smallest in the analysis (N=77). While the effect was statistically significant, future research with a larger sample is needed to confirm the robustness of this specific finding.

5.2 Discussion of Findings for H2: The Market-Driven Path to Exit

While the findings for follow-on funding revealed a complex interplay of investor signals, the analysis of Hypothesis 2 provides a clearer, though still nuanced, illustration of how funding pathways are associated with the likelihood and speed of a commercial exit. The results suggest that while a prevalence of private funding is indeed a strong predictor of an exit, the equally strong performance of hybrid-funded ventures points to a more complex mechanism centered on the influence of market logic.

The Expressway to Exit: The Dominance of Market Logic

The analysis indicates a strong positive association between a funding history dominated by market-oriented investors and a faster, more probable exit. As the purely private pathway was the reference category in the model, its strong performance is inferred from the significantly weaker performance of the public funding path relative to it.

Notably, startups with a high prevalence of hybrid funding were statistically indistinguishable from their privately-funded peers in both their likelihood and speed of exiting. This

finding offers only partial support for the original hypothesis, which singled out private funding. Instead, it suggests that the crucial driver is the presence of market logic within the funding syndicate. For purely private investors, the entire model of engagement is optimized to achieve a liquidity event. For hybrid investors, it appears this market-driven demand for an exit often overrides the state partner's preference for patience. Therefore, the strong correlation between these two pathways and successful exits likely reflects the targeted support provided by investors whose primary goal is a profitable sale or IPO.

The Patient Path: Public Funding's Alignment with State Logic

Conversely, a high proportion of public funding was the strongest statistical predictor against achieving a commercial exit. This association, however, should not necessarily be interpreted as a failure of public funding. It is more likely a direct manifestation of the state logic that guides these investors. With primary goals centered on long-term innovation, regional development, or correcting market failures, a rapid exit can be contrary to the mission. The "patient capital" provided by public entities appears to successfully insulate ventures from market pressures, allowing them to pursue longer-term R&D and impact generation—a different, but equally valid, strategic path.

Finally, beyond the central role of funding composition, the models highlighted other significant predictors of a startup's exit. The finding that the 'age at last round' negatively impacts the likelihood of an exit suggests that momentum is critical; ventures that go longer without securing fresh capital may be perceived as stagnating, reducing their attractiveness for acquisition. Similarly, the positive association between the 'total number of funding rounds' and the hazard of an exit aligns with expectations, as each successful round acts as a step of validation and provides resources that move a startup closer to the scale required for an IPO or a strategic sale. While not the primary focus of this study's theory, these factors provide crucial context, illustrating that the path to an exit is shaped by both the type of funding and the rhythm of the financing process itself.

5.3 Discussion of Findings for H3: The Complexities of Long-Term Survival

Shifting from the clear-cut pathways to an exit, the investigation into Hypothesis 3 — the link between public funding and long-term survival — yielded the most complex results of this study. This complexity stems largely from a critical, overarching methodological limitation established earlier: survivor bias. Because the analysis focuses exclusively on mature firms that have already survived for a decade, it cannot measure the role funding plays in helping a venture reach that milestone in the first place. All subsequent findings must be interpreted through this lens. With this key caveat in mind, the analysis did not support the simple hypothesis that a higher prevalence of public funding increases survival; instead, it points towards the structure of capital being more influential.

The Fading Signal of Patient Capital

The initial regression model showed results contrary to the 'patient capital' hypothesis. It was the amount of private and hybrid funding—not public—that showed a significant positive correlation with endurance past the ten-year mark. This finding may be explained by the 'washed-out' effect: after a decade of operations, the specific advantages of early-stage public funding may fade, becoming less critical than more contemporary factors or other unobserved complexities like team quality and market adaptation, which likely become the dominant drivers of success for mature firms.

The Nuances of Funding Structure

Even within this cohort of survivors, a more granular analysis revealed that the structure of funding can be more important than its source. Public non-dilutive funding (i.e., grants) did emerge as a marginally significant positive predictor of survival, suggesting that for mature firms, "clean" capital that provides stability without complicating governance is still beneficial. The faintness of this statistical signal, however, may itself be a consequence of the survivor bias—a lingering echo of a much stronger, but un-measurable, early-stage survival effect that is obscured by the study's design.

5.4 Holistic Synthesis: The Strategic Roles of Funding in Sustainable Innovation

Having dissected the findings for each hypothesis individually, this final section of the discussion adopts a more holistic perspective. By weaving together the key threads from the analyses of follow-on funding, exits, and long-term survival, it is possible to derive several overarching conclusions about the complex funding landscape for sustainable startups.

Private Funding: The Effectiveness of Market-Driven Support

Across all three success metrics, private funding emerged as a consistently effective pathway. It was associated with securing the largest amounts of follow-on funding, a high likelihood of a swift and profitable exit, and was also a significant predictor of long-term survival. This suggests a strong compatibility between the 'market logic' of private investors and the needs of scalable sustainable startups. The active, hands-on support and commercial discipline provided by these investors appear to be crucial for navigating the path to scale. Furthermore, this strong performance may also reflect a maturation of the investment landscape itself; the growing trend of impact investing has likely equipped many private funds with specialized expertise in evaluating and nurturing ventures that blend profit with purpose.

Public Funding: The Critical Distinction of Structure

In stark contrast, public funding in its general form was not associated with strong performance on the observed market-based metrics, a finding that raises questions about

its effectiveness in catalyzing private investment. However, the most insightful finding of this thesis is that public funding is not monolithic. The analyses consistently revealed a dramatic split based on its structure. Dilutive public funding was repeatedly associated with the least favorable outcomes, likely because it sends a negative signal to the market about a venture's inability to attract private capital on its own terms. Conversely, non-dilutive public funding—such as grants—acted as a powerful and clean certification of quality, putting startups on a trajectory to success equivalent to their privately-backed peers. This distinction is paramount, though the finding for dilutive public funding must be interpreted with caution, given it represented a smaller sample in the dataset.

Hybrid Funding: A Successful Synthesis of Logics

Hybrid funding consistently performed 'toe-to-toe' with purely private funding on all market-oriented outcomes. This success offers two key insights. First, it strongly indicates that when public and private logics are formally blended, it is the market logic that tends to dominate the venture's trajectory towards follow-on funding and exit. Second, this is a powerful endorsement of the public policy goal behind such models. The findings suggest that hybrid structures are working as intended: public capital is successfully leveraged to de-risk opportunities and "crowd-in" the efficiency, expertise, and capital of the private sector, guiding sustainable innovations towards commercial scale.

Universal Truths: The Underlying Physics of Startup Growth

Finally, across all archetypes, the analysis revealed several factors that act as universal laws of startup physics. The findings that the 'age at last round' negatively impacts exit chances and that the total funding amount is positively associated with success highlight truths common to all ventures. Regardless of the type of capital, a startup's journey appears to be governed by momentum, continuous validation, and sufficient capitalization. These factors provide crucial context, reminding us that while the choice of investor sets the strategic direction, a startup must still obey the fundamental rules of growth and market confidence to succeed.

This discussion has woven the study's empirical results into a cohesive narrative, using theory to interpret the distinct strategic pathways associated with private, public, and hybrid funding. With this analytical work complete, the final chapter now turns to summarizing the study's primary contributions and looking forward. It will outline the key theoretical advancements and practical implications for entrepreneurs and policymakers, before formally addressing the study's limitations and suggesting concrete directions for future research.

6 Conclusion

The journey of a sustainable startup is profoundly shaped by the institutional logic of its financial backers. This thesis has demonstrated that the choice between private, public, and hybrid funding is not merely a financial transaction but a foundational strategic decision that sets a venture on a distinct path toward either rapid commercialization or long-term operational endurance. This concluding chapter summarizes the key findings that support this conclusion and outlines their theoretical contributions, practical implications, and directions for future research.

The key findings of this study are:

- **Private Funding: A Strong Enabler of Commercial Success.** The findings consistently showed that private funding is a highly effective pathway for sustainable startups aiming for scale. Once initial funding was secured, a high prevalence of private capital was associated with securing the largest amounts of follow-on funding, achieving a faster and more probable commercial exit, and was also a positive predictor of long-term survival. This suggests a strong compatibility between the market logic and hands-on support of private investors and the needs of growth-oriented sustainable ventures.
- **Public Funding: The Critical Distinction of Structure.** The most striking finding regarding public funding is that its influence is not monolithic but is critically dependent on its structure. The analysis revealed a dramatic split: while dilutive public equity was associated with a significantly lower likelihood of attracting follow-on funding, non-dilutive public grants acted as a powerful, positive certification of quality, performing on par with private and hybrid funding. This suggests that the way public support is delivered is more important than its source alone.
- **Hybrid Funding: A Successful Synthesis Dominated by Market Logic.** Hybrid funding consistently performed on par with purely private funding across all market-oriented metrics, including likelihood of follow-on funding and the probability and speed of an exit. This success offers a dual insight: it strongly indicates that the market logic of the private co-investor tends to dominate the venture's strategic trajectory, while also confirming the effectiveness of the public policy design. Hybrid models appear to be working as intended, successfully using public funds to de-risk opportunities and leverage private sector efficiency to guide sustainable innovations toward commercial scale.

These core findings give rise to several important contributions to both academic theory and professional practice.

6.1 Theoretical Contributions

This thesis contributes to the literature on startup finance by adopting a more holistic theoretical lens than is common in studies focused on singular funding instruments. By applying the theory of Institutional Logics, this research moves beyond a purely financial assessment to differentiate public, private, and hybrid funding archetypes based on their underlying motivations and goals. While this high-level approach takes an aggregated view, its value is demonstrated by the significant and plausible results it delivered regarding the divergent pathways of sustainable startups.

Within this framework, two specific contributions should be highlighted. First, the explicit operationalization of hybrid funding provides a clear model for understanding it as a synthesis of market and state logics, a category often under-examined in the literature. Second, the adoption of a multi-dimensional view of success—assessing follow-on funding, exit, and survival—allowed for a more complete and nuanced analysis by aligning performance metrics with the competing objectives of each funding archetype.

6.2 Practical Implications

Alongside the theoretical contributions, this thesis offers several practical implications for the key actors within the sustainable innovation ecosystem:

For Entrepreneurs

The results provide a clear strategic guide for founders. The hands-on support and commercial discipline from investors with a private background—whether in purely private or hybrid arrangements—appear invaluable for achieving growth and a successful exit. At the same time, entrepreneurs should actively pursue public grants, as this form of non-dilutive funding provides crucial capital with few conditions. Founders should, however, approach dilutive public equity with caution, as the findings suggest this pathway may lack the market-oriented drive of its private counterparts and could send a negative signal to the investment community.

For Policymakers

The same results, when interpreted from a policy perspective, offer clear directives for program design. Non-dilutive grants are generally shown to be an effective instrument for achieving the public good of regional development. However, the findings also suggest that governments may be less effective when running their own direct equity programs, a role perhaps better left to private, incentive-based investors. Instead, the most potent strategy for public intervention appears to be the creation and support of hybrid models. Their strong performance in this study confirms they are an efficient mechanism for using public funds to de-risk opportunities and leverage private sector expertise.

For Investors

For conventional investors, the direct implications are more nuanced, given the study's focus on sustainable startups without a non-sustainable comparison group. However, the findings clearly demonstrate that sustainable ventures are capable of achieving strong, market-based outcomes when backed by the right capital structure. In the broader context of global challenges like climate change, the evidence that these dual-mission startups can succeed offers a compelling, data-driven reason for investors to continue deepening their expertise and engagement in this critical sector.

6.3 Limitations and Directions for Future Research

Before concluding this thesis, it is essential to acknowledge its limitations, which in turn highlight promising directions for future research.

On a theoretical level, this study applied broad concepts like Institutional Logics in a "dif-fused" manner to create its high-level archetypes. Future research could adopt a more granular approach, conducting qualitative case studies to explore the specific nuances of investor motivations in greater depth.

Thoroughly discussed in section 3.6, methodologically, the analysis relies on second-hand data from *Crunchbase* which, while extensive, cannot be fully validated and led to operationalization challenges like the significant survivor bias in the H3 analysis. As also detailed in that section, the study's observational design means the findings represent strong associations rather than strict causal relationships and cannot rule out the influence of unobserved variables like founder quality. A clear avenue for future work would be to validate these quantitative findings using first-hand data from interviews or surveys with founders and investors.

Furthermore, while the thesis revolves around "sustainable" startups, a significant limitation is its reliance on market-based success metrics due to data availability. A crucial next step for this research agenda would be to re-evaluate these funding pathways using direct measures of sustainability performance, such as alignment with the Sustainable Development Goals or quantified environmental impact. This would reveal whether the pathways to commercial success are also the pathways to the greatest societal impact.

Finally, for the sake of focus, this analysis used geography and industry primarily as control variables. A deeper investigation into the differences between specific regions (e.g., the policy and investment environments in Europe vs. North America) or within specific sustainable industry sub-sectors would be a valuable and logical extension of this work.

6.4 Concluding Remarks

This research has focused on the funding pathways of sustainable startups, recognizing their critical role in addressing the immense environmental and societal challenges ahead. It is essential, however, to conclude with a crucial reminder: while important, optimizing

the financial ecosystem for these ventures is no substitute for direct and effective climate policy (Bucher-Koenen et al. 2025). Policies that universally target the real economy can address environmental impacts at their source, whereas the indirect route of financial market regulation is inherently more limited. As this thesis has shown, its impact is contingent on a startup's need for external financing and the specific type of capital it secures. Therefore, fostering this funding ecosystem should be seen as one vital component of a much broader policy effort required to navigate a successful green transition.

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A Appendix

A.1 Overview of the *Crunchbase* Industry Group: Sustainability

Table A.1: Overview of the *Crunchbase* Industry Group: Sustainability

Industry	Definition
Biofuel	Companies that revolve around a fuel which is produced through contemporary processes from biomass rather than a fuel produced by the very slow geological processes involved in the formation of fossil fuels such as oil.
Biomass Energy	Companies that develop the use of organic materials (a renewable and sustainable source of energy) to create electricity or other forms of power.
Clean Energy	Companies involved in creating energy that is produced through means that do not pollute the atmosphere.
CleanTech	Companies that deal with any process, product, or service that reduces negative environmental impacts through significant energy sourcing or energy efficiency improvements, the sustainable use of resources, or environmental protection activities.
Energy Efficiency	Companies that provide products or services that aim to reduce the amount of energy required to perform a task.
Environmental Engineering	Companies that use scientific and engineering strategies to improve the quality of the environment, and to protect the health of living organisms from adverse environmental effects, such as pollution.
Green Building	Companies that apply environmentally responsible and resource-efficient processes throughout a building's life-cycle.
Green Consumer Goods	Companies that produce goods that have been made in a way that protects the natural environment.
GreenTech	Companies that focus on the use of technology or technological processes to reduce negative impacts on the environment, conserve resources, and promote sustainability.
Natural Resources	Companies dealing with materials or substances that occur in nature and can be used for economic gain, such as minerals, forests, water, and fertile land.

Industry	Definition
Organic	Companies that relate to products derived or related to living matter.
Pollution Control	Companies that keep air, water, or other pollution output below specific levels.
Recycling	Companies involved in the process of converting waste materials into new materials and objects.
Renewable Energy	Companies that offer products or services related to the collection of energy from resources that are naturally replenished more quickly than they are consumed, including but not limited to, sunlight, wind, water movement, geothermal heat, and biomass.
Solar	Companies that research, develop, engineer, manufacture, or install technologies to capture radiant light and heat from the sun as a source of renewable energy.
Sustainability	Companies that develop and promote environmentally friendly products, services, and practices that minimize negative impacts on the environment and natural resources, while promoting social and economic well-being.
Waste Management	Companies that process and manage waste, i.e. garbage disposal, refuse disposal, recycling, etc.
Water Purification	Companies that deal with the process of removing undesirable chemicals, biological contaminants, suspended solids, and gasses from water.
Wind Energy	Companies that are part of the wind-making electricity process, from companies that build and maintain wind turbines to ones that store/distribute energy.

Source: This is an excerpt from *Crunchbase's* Glossary of Industries (Crunchbase Inc. 2025b) (accessed in April 2025= for Industries part of the 'Sustainability' Industry Group.

A.2 Classification of Funding Nature

Table A.2: Classification of Funding Nature

Funding Type	Description	Dilutive / Non-dilutive
Angel	Investment from an affluent individual providing capital for a business start-up, usually in exchange for convertible debt or ownership equity.	Dilutive
Convertible Note	A form of short-term debt that converts into equity, typically in a future financing round.	Dilutive
Corporate Round	A funding round led by a corporation, often for strategic purposes, which may involve an equity stake.	Dilutive
Debt Financing	Capital that a business raises by taking out a loan. The debt must be repaid, typically with interest, and does not involve selling equity.	Non-dilutive
Grant	A sum of money given by a government or other organization for a particular purpose, with no expectation of repayment or equity.	Non-dilutive
Non-equity Assistance	Support for a business that does not involve providing capital, such as mentorship, resources, or strategic partnerships.	Non-dilutive
Pre-seed	The earliest stage of funding for a startup, typically used to get the business off the ground in exchange for equity.	Dilutive
Private Equity	Investment from private equity funds that invest in or acquire private companies that are not listed on a public stock exchange.	Dilutive
Seed	The first official equity funding stage. It typically represents the first official money that a business raises.	Dilutive
Series A, B, C...	A series of venture capital funding rounds, typically denoted by letters, that represent stages of maturity and growth in exchange for equity.	Dilutive
Undisclosed	A funding round where the financial details have not been publicly disclosed. It could be dilutive or non-dilutive.	Varies / Undisclosed
Venture - Series Unknown	A venture capital funding round where the specific series (A, B, etc.) is not known, but involves selling equity.	Dilutive

Source: Adapted from *Crunchbase's Glossary of Funding Types* (Crunchbase Inc. 2024). The "Dilutive / Non-dilutive" classification is the author's. (Accessed in May 2025).

A.3 List of Variables

Table A.3: List of Variables

Variable	Description
<i>Dependent Variables</i>	
Follow-On Funding (bin_follow_on_funding)	A binary variable coded as '1' if a startup received any funding after its initial round and '0' otherwise.
Logarithmic Follow-On Amount (log_follow_on_amount)	The log-transformed total U.S. dollar amount of all follow-on funding received by a startup.
Exit Status (bin_exit_status)	A binary variable coded as '1' if a startup experienced an exit (acquisition or IPO) and '0' otherwise.
Time to Exit (time_to_exit_years)	A duration variable measuring the time in years from a startup's founding to its exit or the end of the observation period.
Operating Status 2025 (bin_op_status_2025)	A binary variable indicating if a startup was verified as operational ('1') or inactive/closed ('0') in May 2025.
<i>Independent Variables</i>	
First Round Funding Type	The classified type ('Public', 'Private', or 'Hybrid') of a startup's first institutional funding round.
Funding Round Counts (e.g., num_public_rounds)	The total number of funding rounds a startup received from each of the three archetypes.
Funding Round Ratios (e.g., ratio_private_rounds)	The proportion of a startup's total funding rounds that belong to each of the three archetypes.
Funding Amount by Archetype (e.g., amount_private_funding_m)	The total sum of funding (in millions USD) a startup received from each of the three archetypes.
<i>Control & Moderating Variables</i>	
Funding Nature (Moderator)	A binary variable classifying a funding round as 'dilutive' (equity) or 'non-dilutive' (grant, debt), used to test for interaction effects.
Startup Age (e.g., age_at_first_round)	The age of a startup in years at the time of a specific event (e.g., first round, last round, exit).
Startup Size / Total Funding	The total cumulative funding amount a startup has raised, used as a proxy for its size.
Round Amount (e.g., first_round_amount_m)	The financial size (in millions USD) of a specific funding round, such as the first or the last.
Event Year (e.g., startup_founded_year)	The calendar year in which an event occurred, used to control for macroeconomic conditions and temporal trends.
Geographic Region	A categorical variable for the startup's headquarters location, grouped into broad regional clusters.
Startup Industry Cluster	A categorical variable for the startup's industry, grouped into higher-level clusters.

Variable	Description
Number of Funding Rounds	The total count of a startup's funding rounds, used as a control for its financing maturity.

A.4 Diagnostic Checks: Multicollinearity

This section details the multicollinearity diagnostics performed for the primary regression models using the Variance Inflation Factor (VIF). A common rule of thumb is that a VIF value exceeding 5 or 10 indicates potentially problematic multicollinearity. As shown in the tables below, while some regional variables exhibit moderate collinearity ($VIF > 5$), all predictor variables remain below the conservative threshold of 10, suggesting that multicollinearity is not a significant issue for the model interpretations. The VIF for each model's intercept term was excluded from the tables as it is not used to assess collinearity among predictors.

A.4.1 Follow-On Funding Model

Two models were used to analyze follow-on funding: one predicting the likelihood of receiving it (`bin_follow_on_funding`) and another predicting the amount (`log_follow_on_amount`). Since both models use the exact same set of predictor variables, their VIF scores are identical and are presented in a single table.

```
DV 1: bin_follow_on_funding ~ age_at_first_round +  
first_round_amount_M + first_round_year +  
C(startup_region) + C(startup_industry_cluster) +  
C(first_round_funding_type) + C(first_round_funding_nature)
```

```
DV 2: log_follow_on_amount ~ age_at_first_round +  
first_round_amount_M + first_round_year +  
C(startup_region) + C(startup_industry_cluster) +  
C(first_round_funding_type) + C(first_round_funding_nature)
```

Table A.4: VIF Results for Follow-on Funding Model

Predictor Variable	VIF
<i>Startup Region (Categorical)</i>	
North America	6.43
Western Europe	6.37
Asia Pacific	3.30
South Asia & MENA	2.65
Latin America	1.86
Central & Eastern Europe (CEE)	1.81
<i>First Round Funding Type (Categorical)</i>	
Public	2.37
Private	1.84
<i>First Round Funding Nature (Categorical)</i>	
Non-dilutive	1.81
<i>Startup Industry Cluster (Categorical)</i>	
Sustainable Energy & Power	1.73
General Sustainability / Other	1.39
Sustainable Manufacturing & Industrial	1.36
Sustainable Digital & AI Solutions	1.22
Sustainable Ag & Food Systems	1.21
Sustainable Mobility	1.12
<i>Continuous Variables</i>	
Age at First Round (Years)	1.76
First Round Year	1.75
First Round Funding Amount (\$M)	1.06

A.4.2 Exit Model

The VIF diagnostics for the Cox Proportional Hazards model on startup exits are presented below.

```
bin_exit_status ~ C(startup_region) + C(startup_industry_cluster) +
startup_founded_date + total_funding_amount_m +
age_at_last_round_years + last_round_funding_amount_m +
ratio_hybrid_rounds + ratio_public_rounds + number_of_funding_rounds
```

Table A.5: VIF Results for Exit Model

Predictor Variable	VIF
<i>Startup Region (Categorical)</i>	
North America	6.47
Western Europe	6.36
Asia Pacific	3.29
South Asia & MENA	2.63
Latin America	1.83
Central & Eastern Europe (CEE)	1.80
<i>Startup Founded Year (Categorical Dummies)</i>	
Year: 2018	2.33
Year: 2017	2.23
Year: 2016	2.20
Year: 2019	2.12
Year: 2013	2.07
Year: 2015	2.01
Year: 2014	1.99
Year: 2012	1.96
Year: 2011	1.81
<i>Startup Industry Cluster (Categorical)</i>	
Sustainable Energy & Power	1.74
General Sustainability / Other	1.40
Sustainable Manufacturing & Industrial	1.37
Sustainable Digital & AI Solutions	1.24
Sustainable Ag & Food Systems	1.23
Sustainable Mobility	1.13
<i>Continuous/Count Variables</i>	
Total Funding Amount (\$M)	1.79
Last Round Funding Amount (\$M)	1.78
Age at Last Round (Years)	1.27
Ratio Hybrid Rounds	1.21
Ratio Public Rounds	1.19
Number of Funding Rounds	1.17

A.4.3 Survival Models

For the survival analysis, VIF was checked for two specifications: a baseline model with aggregated funding amounts, and a second model where these amounts were disaggregated by their dilutive nature. The results are presented side-by-side to assess multicollinearity in both models.

Model 1 (Aggregated Funding):

$\text{bin_op_status_2025} \sim C(\text{startup_region}) + C(\text{startup_industry_cluster}) +$

startup_founded_year + number_of_funding_rounds +
hybrid_funding_amount_m + private_funding_amount_m +
public_funding_amount_m

Model 2 (Disaggregated Funding):

bin_op_status_2025 ~ C(startup_region) + C(startup_industry_cluster) +
startup_founded_year + number_of_funding_rounds +
hybrid_dilutive_funding_amount_m + private_dilutive_funding_amount_m +
public_dilutive_funding_amount_m + hybrid_non_dilutive_funding_amount_m +
private_non_dilutive_funding_amount_m + public_non_dilutive_funding_amount_m

Table A.6: Side-by-Side VIF Results for Survival Models

Predictor Variable	VIF (Model 2)	VIF (Model 3)
<i>Startup Region (Categorical)</i>		
Western Europe	6.80	6.81
North America	6.63	6.63
Asia Pacific	3.21	3.21
South Asia & MENA	2.65	2.65
Latin America	1.89	1.89
Central & Eastern Europe (CEE)	1.66	1.66
<i>Startup Industry Cluster (Categorical)</i>		
Sustainable Energy & Power	1.81	1.81
Sustainable Manufacturing & Industrial	1.40	1.40
General Sustainability / Other	1.39	1.40
Sustainable Ag & Food Systems	1.24	1.24
Sustainable Digital & AI Solutions	1.21	1.21
Sustainable Mobility	1.11	1.11
<i>Funding Amount Variables</i>		
Hybrid Funding Amount (\$M)	1.01	–
Private Funding Amount (\$M)	1.06	–
Public Funding Amount (\$M)	1.02	–
Hybrid Dilutive Amount (\$M)	–	1.03
Private Dilutive Amount (\$M)	–	1.08
Public Dilutive Amount (\$M)	–	1.06
Hybrid Non-dilutive Amount (\$M)	–	1.00
Private Non-dilutive Amount (\$M)	–	1.06
Public Non-dilutive Amount (\$M)	–	1.01
<i>Other Continuous/Count Variables</i>		
Number of Funding Rounds	1.07	1.10
Startup Founded Year	1.03	1.04

A.5 Supplementary Findings

Table A.7: Nested Logistic Regression Results for Long-Term Survival

Variable	Model 1 (Controls)		Model 2 (+ Archetype)		Model 3 (+ Archetype & Nature)	
	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.	Coef. (OR)	Std. Err.
Funding Variables						
Total Funding Amount (\$M)	0.0031** (1.003)	(0.001)				
Private Funding Amount (\$M)			0.0025** (1.003)	(0.001)		
Public Funding Amount (\$M)			0.0296 (1.030)	(0.020)		
Hybrid Funding Amount (\$M)			0.1506** (1.163)	(0.068)		
Private Dilutive Amount (\$M)					0.0023* (1.002)	(0.001)
Public Dilutive Amount (\$M)					0.0054 (1.005)	(0.013)
Hybrid Dilutive Amount (\$M)					0.1319** (1.141)	(0.067)
Private Non-Dilutive Amount (\$M)					0.0050 (1.005)	(0.005)
Public Non-Dilutive Amount (\$M)					0.0924* (1.097)	(0.053)
Hybrid Non-Dilutive Amount (\$M)					5.4593 (234.9)	(4.149)
Control Variables						
Number of Funding Rounds	0.0820 (1.085)	(0.067)	0.0325 (1.033)	(0.068)	0.0123 (1.012)	(0.069)
Founding Year	-0.0601 (0.942)	(0.048)	-0.0557 (0.946)	(0.049)	-0.0578 (0.944)	(0.049)
Industry Dummies	Yes		Yes		Yes	
Region Dummies	Yes		Yes		Yes	
Intercept	122.2261	(97.351)	113.1173	(98.135)	117.1498	(98.776)
Model Fit						
Observations	905		905		905	
Log-Likelihood	-483.28		-475.61		-470.54	
Pseudo R ²	0.036		0.052		0.062	

Note: Coefficients are log-odds with Odds Ratios (OR) in parentheses below. Standard errors are in parentheses next to coefficients. Regional and industry cluster dummy variables were included but are not shown for brevity. Significance levels: * p<0.1, ** p<0.05, *** p<0.01.

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